



# The broken adaptive ridge estimation for the high-dimensional covariate-adjusted regression model

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Received: 11 November 2025 / Accepted: 17 March 2026  
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## Abstract

We consider the covariate-adjusted regression (CAR) model in which both predictors and response are not observable but distorted by a common confounding variable in multiplication way. In this paper, we employ the broken adaptive ridge (BAR) method to simultaneously select variables and estimate the coefficients for high-dimensional CAR model and obtain a CAR-BAR (CBAR) estimator after the predictors and response adjusted. We establish that CBAR estimator is consistent for variable selection and oracle for parameter estimation and possesses a grouping property for highly correlated covariates. Further, a consistent estimator of the error variance is given. Extensive simulations indicate that our CBAR method is superior to LASSO, Adaptive LASSO, SCAD and MCP methods. Particularly, a simulation scenario involving collinear predictors also demonstrates that CBAR outperforms other methods. Moreover, we show that the CBAR enjoys fast convergence rate by a straightforward simulation. In the analysis of Boston Housing data, we construct a statistic to test whether the selected variables have significant effect on the response, and give different and interesting explanations for the fitted model.

**Keywords** Covariate-adjusted regression model · Grouping effect · Parameter estimation · Oracle estimator · Variable selection

## 1 Introduction

Measurement error data often arise in some areas such as biomedical studies and health-related researches. Among them, a common data type is that the response and predictors are distorted by an observable confounding covariate in a multiplicative

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fashion. The model that characterizes this type data is known as covariate-adjusted regression model (CAR) which is initially proposed by (Şentürk and Müller, 2005a, b). The CAR model is

$$\begin{cases} Y_i = \sum_{j=1}^p X_{ij}\beta_j + \varepsilon_i, \\ \tilde{Y}_i = \psi(U_i)Y_i, \\ \tilde{X}_{ij} = \phi_j(U_i)X_{ij}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, p, \end{cases} \quad (1)$$

where  $Y_i$ ,  $i = 1, 2, \dots, n$  are unobservable responses,  $X_{ij}$ ,  $i = 1, 2, \dots, n$ ,  $j = 1, 2, \dots, p$  are unobservable values of predictors,  $\beta = (\beta_1, \beta_2, \dots, \beta_p)^T$  is the unknown vector of regression coefficients, the error  $\varepsilon_i$  satisfies that  $E(\varepsilon_i) = 0$  and  $Var(\varepsilon_i) = \sigma^2 < \infty$ ,  $i = 1, 2, \dots, n$ , and  $\psi(\cdot)$ ,  $\phi_j(\cdot)$ ,  $j = 1, 2, \dots, p$  are unknown distorting functions of an observable confounding variable  $U$ . But,  $(\tilde{Y}_i, \tilde{X}_{i1}, \tilde{X}_{i2}, \dots, \tilde{X}_{ip}, U_i)$ ,  $i = 1, 2, \dots, n$  are independent and identically distributed observations of the actual observable response, predictors and confounding variable. Then, parameter estimators need to be provided based on these observation results. If we regress between actual observable response and predictors without considering the distortion, the estimators of parameters may be non negligibly biased and inconsistent. As a remedy, Şentürk and Müller (2006) introduced a flexible multiplicative adjustment and investigated the asymptotic normality of the estimators for covariate-adjusted linear model. Later on, there have been much literature on the statistical modeling of such distorted data. Şentürk and Müller (2009) considered transformation methods for covariate-adjusted generalized linear models. Cui et al. (2009) proposed the estimation of the distorting functions by non-parametric regression of the predictors and response on the distorting covariate to recover  $Y_i$  and  $X_{ij}$ ,  $i = 1, 2, \dots, n$ ,  $j = 1, 2, \dots, p$ . Thus, these data recovery method can be applied to various CAR models, (such as (Li et al., 2010; Zhang et al., 2012a, b, 2013; Du et al., 2015; Zhang et al., 2015, 2019; Pan et al., 2023)). As a surrogate to  $L_0$  penalty, Dai et al. (2018) proposed the re-weighted  $L_2$ -penalization named as broken adaptive ridge (BAR) method. The BAR procedure enjoys not only the oracle property but also the grouping effect for highly correlated covariates, a desirable feature not shared by most other variable selection procedures. Due to its appealing properties, the BAR procedure has been extended to various models (see, e.g., (Zhao et al., 2020; Kawaguchi et al., 2020; Du et al., 2022; Zhao et al., 2024; Mahmoudi and Lu, 2025)).

This article uses the BAR method to simultaneously estimate parameters and select variables for CAR model (1). By regressing the contaminated response and predictors on the common confounder as (Cui et al., 2009), we first obtain consistent estimators of the distorting function  $\psi(\cdot)$  and  $\phi_r(\cdot)$ ,  $r = 1, 2, \dots, p$ , respectively. Second we obtain the estimates of the real response and predictors, respectively. Third we apply the BAR method to estimate  $\beta$  based on recovered data of  $Y_i$  and  $X_{ij}$ ,  $i = 1, 2, \dots, n$ ,  $j = 1, 2, \dots, p$ , and the estimator is named as CBAR in the paper. Finally, the error variance  $\sigma^2$  is consistently estimated. In theory, we have demonstrated that, for divergent dimensionality  $p$ , the proposed CBAR estimator is

consistent in variable selection and has the oracle property. Particularly, for highly correlated covariates, the CBAR estimator possesses a grouping property. Simulations show that the CBAR procedure outperforms other four methods in both variable selection and estimation.

The rest of the paper is organized as follows. In Section 2, we propose the CBAR procedure. In Section 3, we establish its oracle property and grouping effect. Section 4 presents simulations to compare our CBAR procedure with some popular  $L_1$  penalization ones. In Section 5, the CBAR procedure is applied to deal with Boston Housing data. Conclusion is given in Section 6. Technical proofs are provided in the Appendix.

## 2 Modeling and estimation

In this section we estimate the parameters in model (1). For clarity, let

$$Y = Y_1, \tilde{Y} = \tilde{Y}_1, U = U_1, \varepsilon = \varepsilon_1, X_j = X_{1j} \text{ and } \tilde{X}_j = \tilde{X}_{1j}, j = 1, 2, \dots, p. \quad (2)$$

For distorting functions and above variables, we assume that (Şentürk and Müller, 2005b; Cui et al., 2009),

- (a)  $E\{\psi(U)\} = 1$  and  $E\{\phi_j(U)\} = 1, j = 1, 2, \dots, p,$
- (b)  $(X_j, U, \varepsilon)$  are mutually independent for  $j = 1, 2, \dots, p,$  and  $(Y, U)$  are independent.

Notice that, combined with the model (1), assumptions (a) and (b) imply that

$$\psi(U) = \frac{E(\tilde{Y}|U)}{EY}, \quad \phi_j(U) = \frac{E(\tilde{X}_j|U)}{EX_j}, \quad j = 1, 2, \dots, p, \quad (3)$$

$$E\tilde{Y} = EY, \quad E(\tilde{X}_j) = EX_j, \quad j = 1, 2, \dots, p. \quad (4)$$

The available data  $(\tilde{Y}_i, \tilde{\mathbf{X}}_i, U_i), i = 1, 2, \dots, n$  are independent and identically distributed sample given by the model (1) with  $\tilde{\mathbf{X}}_i = (\tilde{X}_{i1}, \tilde{X}_{i2}, \dots, \tilde{X}_{ip})$ . We first give the kernel estimators of distorting functions. Denote by  $f(u)$  the density function of  $U$ , and

$$g_Y(u) = E(\tilde{Y}|u)f(u), \quad g_j(u) = E(\tilde{X}_j|u)f(u), \quad j = 1, 2, \dots, p. \quad (5)$$

Define functional estimators

$$\begin{aligned}\hat{f}(u) &= \frac{1}{nh} \sum_{i=1}^n K\left(\frac{u - U_i}{h}\right), \quad \hat{g}_Y(u) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{u - U_i}{h}\right) \tilde{Y}_i, \\ \hat{g}_j(u) &= \frac{1}{nh} \sum_{i=1}^n K\left(\frac{u - U_i}{h}\right) \tilde{X}_{ij}, \quad j = 1, 2, \dots, p,\end{aligned}\quad (6)$$

where  $K(\cdot)$  is a kernel function and  $h$  is the bandwidth. We know that  $\hat{f}(u)$  is a density estimator of  $f(u)$ . From (3) to (6), we give the estimators of  $\psi(\cdot)$  and  $\phi_j(\cdot)$  as

$$\hat{\psi}(u) = \frac{\hat{g}_Y(u)}{\hat{f}(u)} \times \frac{1}{\bar{Y}}, \quad \hat{\phi}_j(u) = \frac{\hat{g}_j(u)}{\hat{f}(u)} \times \frac{1}{\bar{X}_j}, \quad j = 1, 2, \dots, p, \quad (7)$$

where  $\bar{Y} = n^{-1} \sum_{i=1}^n \tilde{Y}_i$ ,  $\bar{X}_j = n^{-1} \sum_{i=1}^n \tilde{X}_{ij}$ . Then we can recover the unobservable response and predictors, respectively (Cui et al., 2009). Denote by  $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^T$  the unobservable response vector, and by  $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_p)$  the  $n \times p$  unobservable predictive matrix with  $\mathbf{X}_j = (X_{1j}, X_{2j}, \dots, X_{nj})^T$ ,  $j = 1, 2, \dots, p$ . The estimates of  $\mathbf{Y}$  and  $\mathbf{X}$  are given by  $\hat{\mathbf{Y}} = (\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n)^T$  and  $\hat{\mathbf{X}} = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \dots, \hat{\mathbf{X}}_p)$ , respectively, where  $\hat{\mathbf{X}}_j = (\hat{X}_{1j}, \hat{X}_{2j}, \dots, \hat{X}_{nj})^T$ , and

$$\hat{Y}_i = \frac{\tilde{Y}_i}{\hat{\psi}(U_i)}, \quad \hat{X}_{ij} = \frac{\tilde{X}_{ij}}{\hat{\phi}_j(U_i)}, \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, p.$$

They are also named as recovered data.

In terms of the recovered data, we can approximate the linear model  $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ ,  $\varepsilon = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)^T$ , by  $\hat{\mathbf{Y}} \approx \hat{\mathbf{X}}\beta + \varepsilon$ . Then we can obtain the estimator of regression coefficient vector  $\beta$  such as the ordinary least squares (OLS) estimator,

$$\hat{\beta}_{ols} = (\hat{\mathbf{X}}^T \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^T \hat{\mathbf{Y}}. \quad (8)$$

However, due to their dependence on a common confounding variable  $U$ , the recovered predictors may have higher correlation. Hence, the ridge estimator is appropriate and defined as

$$\hat{\beta}_{ridge}(\xi) = (\hat{\mathbf{X}}^T \hat{\mathbf{X}} + \xi \mathbf{I})^{-1} \hat{\mathbf{X}}^T \hat{\mathbf{Y}}, \quad (9)$$

where  $\xi > 0$  is a ridge parameter. Further if the vector  $\beta$  is sparse, its estimate should have many components equal 0. Therefore, we can obtain the sparse estimator of  $\beta$  as follows. Minimize the following penalized least-squares function (Dai et al., 2018)

$$\|\hat{\mathbf{Y}} - \hat{\mathbf{X}}\beta\|^2 + \lambda \sum_{j=1}^p \frac{\beta_j^2}{\hat{\beta}_j^2},$$

where  $\|\cdot\|$  represents the Euclidean norm of a vector,  $\lambda > 0$  is a tuning parameter and  $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p)^T$  is an estimator given in advance. Iteratively define

$$\hat{\beta}^{(k)} = \arg \min_{\beta} \left\{ \left\| \hat{\mathbf{Y}} - \hat{\mathbf{X}}\beta \right\|^2 + \lambda \sum_{j=1}^p \frac{\beta_j^2}{\left( \hat{\beta}_j^{(k-1)} \right)^2} \right\}, \quad k = 1, 2, \dots \quad (10)$$

where for  $k > 1$ ,  $\hat{\beta}^{(k-1)}$  is an adaptive reweighted  $L_2$  penalization estimator obtained from the previous iteration and for  $k = 1$ , the initial estimator  $\hat{\beta}^{(0)}$  is the ridge estimator. Denote by  $\mathbf{D}(\hat{\beta}) = \text{diag}(\hat{\beta}_1^2, \hat{\beta}_2^2, \dots, \hat{\beta}_p^2)$  the matrix whose diagonal elements are the squares of components of  $\hat{\beta}$  and off-diagonals are zero. We can explicitly represent the estimator in (10) as

$$\hat{\beta}^{(k)} = \left( \hat{\mathbf{X}}^T \hat{\mathbf{X}} + \lambda \mathbf{D}^{-1}(\hat{\beta}^{(k-1)}) \right)^{-1} \hat{\mathbf{X}}^T \hat{\mathbf{Y}}. \quad (11)$$

If the initial  $\hat{\beta}^{(0)} = \hat{\beta}_{\text{ridge}}(\xi)$ , which is the ridge estimator given in (9), then the covariate-adjusted broken adaptive ridge (CBAR) estimator of  $\beta$  is defined as

$$\hat{\beta}^* = \lim_{k \rightarrow \infty} \hat{\beta}^{(k)}. \quad (12)$$

**Remark 1** We give some explanations of above definition of the CBAR estimator.

(a). If  $k = 1$ , the estimator  $\hat{\beta}^{(0)} = (\hat{\beta}_1^{(0)}, \hat{\beta}_2^{(0)}, \dots, \hat{\beta}_p^{(0)})^T$  on the right side of (10) is the ridge estimator given in (9).

(b). We can rewrite estimator in (11) as

$$\begin{aligned} \hat{\beta}^{(k)} &= \left( \hat{\mathbf{X}}^T \hat{\mathbf{X}} + \lambda \mathbf{D}^{-1}(\hat{\beta}^{(k-1)}) \right)^{-1} \hat{\mathbf{X}}^T \hat{\mathbf{Y}} \\ &= \left( \mathbf{D}(\hat{\beta}^{(k-1)}) \hat{\mathbf{X}}^T \hat{\mathbf{X}} + \lambda \mathbf{I}_p \right)^{-1} \mathbf{D}(\hat{\beta}^{(k-1)}) \hat{\mathbf{X}}^T \hat{\mathbf{Y}} \\ &= \left( \left( \hat{\beta}_1^{(k-1)} \hat{\mathbf{X}}_1, \dots, \hat{\beta}_p^{(k-1)} \hat{\mathbf{X}}_p \right)^T \hat{\mathbf{X}} + \lambda \mathbf{I}_p \right)^{-1} \left( \hat{\beta}_1^{(k-1)} \hat{\mathbf{X}}_1, \dots, \hat{\beta}_p^{(k-1)} \hat{\mathbf{X}}_p \right)^T \hat{\mathbf{Y}}, \end{aligned}$$

where  $\hat{\beta}^{(k-1)} = (\hat{\beta}_1^{(k-1)}, \dots, \hat{\beta}_p^{(k-1)})^T$  for  $k = 1, 2, \dots$ . Hence, the estimators given in (10) or (11) for all  $k \geq 1$  are well-defined.

(c). The estimator  $\hat{\beta}^{(k)}$  can be thought as  $k$  order adaptive calibration of ridge estimator  $\hat{\beta}^{(0)}$ . Although  $\hat{\beta}^*$  is defined as a limit, the adaptive calibration will be broken at order  $k$  if  $\hat{\beta}^{(k)}$  and  $\hat{\beta}^{(k-1)}$  are very close. When the condition  $\|\hat{\beta}^{(k)} - \hat{\beta}^{(k-1)}\| \leq \epsilon$  (set to  $10^{-5}$ ) is reached, the iterative algorithm is stopped. Then the estimator  $\hat{\beta}^{(k)}$  is approximately taken to be the CBAR estimator.

(d). Unlike the usual linear regression, the response vector  $\hat{\mathbf{Y}}$  and predictor matrix  $\hat{\mathbf{X}}$  are adjusted by the covariate  $U$ . Hence, we call  $\hat{\beta}^*$  the covariate-adjusted broken adaptive ridge (CBAR) estimator.

**Remark 2** In the standard linear model, there are no distortions with  $\psi(\cdot) = 1$  and  $\phi_r(\cdot) = 1$ ,  $r = 1, 2, \dots, p$ , and both  $\mathbf{X}$  and  $\mathbf{Y}$  are directly observable. In the asymptotic normality derivation of Dai et al. (2018), their equation (A.25) includes a sum of i.i.d. error terms  $\varepsilon_i$ . Consequently, by applying the Lindeberg–Feller central limit theorem directly, (A.25) converges in distribution to  $N(0, 1)$ . In CAR setting, however, the situation is more complex. We use  $U$ -statistics theory to establish the relationships between recovered data  $(\hat{\mathbf{Y}}, \hat{\mathbf{X}})$  and real unobservable variables, which are used to obtain asymptotic normality of CBAR estimator. Because our derivation of asymptotic normality is based on the  $(\hat{\mathbf{Y}}, \hat{\mathbf{X}})$ , the final expression (A.9) includes two additional, correlated terms arising from the estimation error. Using  $U$ -statistics techniques, we show that the two additional terms are the sum of independent and identically distributed variables, respectively. Finally, we calculate the asymptotic variance of the CBAR estimator, which includes an extra term owing to the distortion; see Remark 4 for details.

Now we investigate the properties of CBAR estimator  $\hat{\beta}^*$  in the high dimension setting where the number of predictors varies with sample size and is denoted by  $p_n$ . Let  $\beta_{0n} = (\beta_{01}, \beta_{02}, \dots, \beta_{0p_n})^T$  be the true parameter value. Without loss of generality, denote  $\beta_{0n} = (\beta_{01}^T, \beta_{02}^T)^T$ , where  $\beta_{01}$  consists of all  $q_n$  nonzero components and  $\beta_{02}$  the remaining zeros. We give the following conditions.

(C1) The functions  $\psi(u)$ ,  $\phi_j(u)$ ,  $j = 1, 2, \dots, p_n$ ,  $f(u)$ ,  $g_Y(u)$  and  $g_j(u)$  are greater than a positive constant and three times differentiable.  $E(\psi(U))^2 < \infty$  and  $E(\phi_j(U))^2 < \infty$ ,  $j = 1, 2, \dots, p_n$ . There exist constants  $\Delta > 0$  and  $c > 0$  such that for any  $|\delta| \leq \Delta$ ,

$$\begin{aligned} |g_r^{(3)}(u + \delta) - g_r^{(3)}(u)| &\leq c|\delta|, \quad 1 \leq r \leq p_n, \\ |g_Y^{(3)}(u + \delta) - g_Y^{(3)}(u)| &\leq c|\delta|, \\ |f^{(3)}(u + \delta) - f^{(3)}(u)| &\leq c|\delta|. \end{aligned}$$

(C2) The continuous kernel function  $K(\cdot)$  is symmetric about zero, supported on  $[-1, 1]$  and have bounded derivatives. Moreover,  $\int_{-1}^1 u^j K(u) du = 0$ ,

$$j = 1, 2, 3.$$

(C3) As  $n \rightarrow \infty$ , the bandwidth  $h$  is chosen to satisfy  $\frac{\log n}{n^{1/4}h} \rightarrow 0$  and  $n^{1/8}h \rightarrow 0$ .

(C4)  $EY$  and  $EX_j$ ,  $j = 1, 2, \dots, p_n$  are bounded away from zero, where  $Y$  and  $X_j$ ,  $j = 1, 2, \dots, p_n$  are given in (2).

(C5) There exists a constant  $M < \infty$  such that  $E(\varepsilon^4) \leq M$  and  $E(X_j^4) \leq M$ ,  $j = 1, 2, \dots, p$ .

- (C6) The matrix  $\Omega = (\omega_{ij})$  with  $\omega_{ij} = E(X_i X_j)$ ,  $i, j = 1, 2, \dots, p_n$  is positively definite, and there exists a constant  $C > 1$  such that  $1/C < \lambda_{\min}(\Omega) \leq \lambda_{\max}(\Omega) < C$ .
- (C7) Let  $a_{0n} = \min_{1 \leq j \leq q_n} |\beta_{0j}|$  and  $a_{1n} = \max_{1 \leq j \leq q_n} |\beta_{0j}|$ . As  $n \rightarrow \infty$ ,  $\xi_n/\sqrt{n} \rightarrow 0$ ,  $\lambda_n a_{1n} q_n^{1/2} a_{0n}^{-2} n^{-1/2} \rightarrow 0$ ,  $a_{1n} q_n^{1/2} p_n^{-1/2} \rightarrow 0$ ,  $a_{1n} q_n^{1/2} n^{-1/2} h^{-2} \log^2 n \rightarrow 0$  and  $a_{1n}^2 p_n^2 n^{-1/2} h^{-1} \log n \rightarrow 0$ , where the subscripts of  $\xi_n$  and  $\lambda_n$  indicate their dependence on  $n$ .

**Remark 3** Condition (C1) is the smoothness requirement of the functions  $f(\cdot)$ ,  $g_Y(\cdot)$  and  $g_r(\cdot)$ ,  $r = 1, 2, \dots, p_n$  (Zhu and Fang, 1996). Conditions (C2)–(C3) are commonly assumed for the  $\sqrt{n}$ -consistency of kernel based estimation (Zhu and Fang, 1996; Cui et al., 2009). Condition (C4) is to ensure that the estimators of the distortion functions are well-defined (Şentürk and Müller, 2006). Condition (C5) is widely assumed to facilitate the technical derivations (Zhu and Zhu, 2009). Conditions (C6) and (C7) are used to derive the oracle property of the CBAR estimator (Dai et al., 2018). Notice that condition (C7) gives some sufficient, but not necessary, conditions needed to prove the numerical convergence and asymptotic properties of the CBAR estimator.

We can give the convergence rates of OLS and ridge estimators as follows.

**Proposition 1** Under conditions (a), (b) and (C1)–(C7), we have

$$\|\hat{\beta}_{ols} - \beta_{0n}\| = O_p(\sqrt{p_n^2/n})$$

where  $\hat{\beta}_{ols}$  is the ordinary least squares estimator given in (8).

**Proposition 2** Assume that the conditions (a), (b) and (C1)–(C7) hold. We have

$$\|\hat{\beta}_{ridge} - \beta_{0n}\| = O_p(\sqrt{p_n^2/n}), \quad (13)$$

where  $\hat{\beta}_{ridge}$  is the ridge estimator given in (9).

Although both  $\hat{\beta}_{ols}$  and  $\hat{\beta}_{ridge}$  are consistent, they can not fulfill the goal of variable selection. We begin to study the CBAR estimator  $\hat{\beta}^*$ . Corresponding the true parameter  $\beta_{0n} = (\beta_{01}^T, \beta_{02}^T)^T$ , we make the following partitions

$$\hat{\beta}^* = (\hat{\beta}_1^{*T}, \hat{\beta}_2^{*T})^T, \quad \mathbf{X} = (\mathbf{X}_I, \mathbf{X}_{II}), \quad \hat{\mathbf{X}} = (\hat{\mathbf{X}}_I, \hat{\mathbf{X}}_{II}),$$

where  $\mathbf{X}_I = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{q_n})$  and  $\hat{\mathbf{X}}_I = (\hat{\mathbf{X}}_1, \hat{\mathbf{X}}_2, \dots, \hat{\mathbf{X}}_{q_n})$ . Denote  $\Omega_1 = (E(X_i X_j))_{q_n \times q_n}$ , the  $q_n$ -order principal submatrix of  $\Omega$  given in the condition (C6), and  $\Gamma_n = (\Gamma_n(i, j))_{q_n \times q_n}$  with

$$\Gamma_n(i, j) = \left\{ \frac{EY^2}{(EY)^2} \text{Var}(\psi(U)) + \frac{E(X_i X_j)}{EX_i EX_j} \text{Cov}(\phi_i(U), \phi_j(U)) \right. \\ \left. - \frac{EY X_i}{EY EX_i} \text{Cov}(\phi_i(U), \psi(U)) - \frac{EY X_j}{EY EX_j} \text{Cov}(\phi_j(U), \psi(U)) \right\} \beta_i \beta_j,$$

for  $1 \leq i, j \leq q_n$ .

**Theorem 1** (Oracle property) Under assumptions (a) and (b), and conditions (C1)-(C7), we have that

- (i) with probability tending to 1, the CBAR estimator  $\hat{\beta}^* = (\hat{\beta}_1^{*T}, \hat{\beta}_2^{*T})^T$  uniquely exists with  $\hat{\beta}_2^* = 0$ , and  $\hat{\beta}_1^*$  is the unique fixed-point of

$$g(\beta_1) = \left\{ \hat{\mathbf{X}}_I^T \hat{\mathbf{X}}_I + \lambda_n \mathbf{D}_1^{-1}(\beta_1) \right\}^{-1} \hat{\mathbf{X}}_I^T \hat{\mathbf{Y}}$$

where  $\mathbf{D}_1(\beta_1) = \text{diag}(\beta_1^2, \beta_2^2, \dots, \beta_{q_n}^2)$ , and

- (ii)  $\sqrt{n} s_n^{-1} \mathbf{b}_n^T (\hat{\beta}_1^* - \beta_{01}) \xrightarrow{D} N(0, 1)$ , where  $\mathbf{b}_n$  is any  $q_n$ -dimension vector satisfying  $0 < \|\mathbf{b}_n\| \leq 1$ , and  $s_n^2 = \mathbf{b}_n^T (\sigma^2 \Omega_1^{-1} + \Gamma_n) \mathbf{b}_n$ .

**Remark 4** In the asymptotic variance  $\sigma^2 \Omega_1^{-1} + \Gamma_n$ , the first term  $\sigma^2 \Omega_1^{-1}$  is the asymptotic variance of BAR estimator (Dai et al., 2018) for the standard linear model with the data are exactly observed without distortion, i.e.,  $\psi(\cdot) = 1$  and  $\phi_j(\cdot) = 1, j = 1, 2, \dots, p_n$ . The extra term  $\Gamma_n$  is due to the multiplicative distortion, which is induced by the unknown functions of the observable covariate  $U$ .

**Remark 5** (Connections with the measurement error models) The CAR framework has connections with measurement error models. In CAR, a natural assumption is that the mean distorting effect should correspond to no distortion (Assumption (a)), It is analogous to the classical measurement errors:  $E(e) = 0$  for  $W = X + e$ , where  $W$  is error-prone,  $X$  is error-free and  $e$  is the measurement error. The CAR model of interest in (1) involves latent variables that cannot be observed directly, in measurement error models,  $X$  is also not directly observable, but real variable.

However, the CAR framework is different from the classical measurement error models. An important difference is that, in the former, the distortions affecting both predictors and responses are the unknown functions of an observable covariate  $U$ . Because  $U$  is observable, this information can be incorporated into the estimation methods, resulting in procedures that differ from those used in measurement error models. For example, the multiplicative distortion functions  $\psi(\cdot)$  and  $\phi(\cdot)$  can be estimated by the kernel smoothing methods, local linear estimation or the spline methods. However, in the literature on measurement error models, most existing methods rely on the normality assumption for the unobserved covariates  $X$  and measurement error  $e$ . In the process of modeling, a key issue is how to eliminate the attenuation bias caused by variance of  $e$ . Several methods can be used to address this, such as,

the attenuation reduction method (Fuller, 1987), the simulation extrapolation method and instrumental variable methods (Carroll et al., 2006).

In some cases, a group of predictors exhibits high correlation and may have similar regression coefficients. We can provide the grouping effect property for CBAR estimator.

**Theorem 2** (Grouping effect) *If assumptions (a) and (b), and conditions (C1)-(C7) hold, then with probability tending to 1 as  $n \rightarrow \infty$ , the CBAR estimator  $\hat{\beta}^* = (\hat{\beta}_1^*, \hat{\beta}_2^*, \dots, \hat{\beta}_{p_n}^*)^T$  satisfies*

$$\left| \frac{1}{\hat{\beta}_i^*} - \frac{1}{\hat{\beta}_j^*} \right| \leq \frac{1}{\lambda_n} \|\hat{\mathbf{Y}}\| \sqrt{2(1 - \rho_{ij})},$$

for nonzero  $\hat{\beta}_i^*$  and  $\hat{\beta}_j^*$ , where  $\rho_{ij}$  denotes the sample correlation coefficient constructed by the recovered data  $\hat{\mathbf{X}}_i$  and  $\hat{\mathbf{X}}_j$ .

The following theorem shows that the estimator  $\hat{\sigma}^2$  obtained by CBAR method is a consistent estimator of the true error variance.

**Theorem 3** *Under the assumptions (a) and (b), and conditions (C1)-(C7), the estimator  $\hat{\sigma}^2 = n^{-1}(\hat{\mathbf{Y}} - \hat{\mathbf{X}}\hat{\beta}^*)^T(\hat{\mathbf{Y}} - \hat{\mathbf{X}}\hat{\beta}^*)$  of  $\sigma^2$  is consistent, that is,  $\hat{\sigma}^2 \rightarrow \sigma^2$  in probability.*

To implement the CBAR procedure, one needs to choose bandwidths  $h_j$  ( $j = 0, 1, \dots, p$ ) and the tuning parameter  $\lambda$  as well as the ridge  $\xi$ . First, for the determination of the bandwidth parameters, we perform the widely used generalized cross-validation (GCV) criterion (Golub et al., 1979). From model (1), we can write  $\hat{Y}_i = \psi(U_i)EY + \psi(U_i)(Y_i - EY)$ ,  $i = 1, 2, \dots, n$ . By the assumption (b), we have

$$E(\psi(U_i)(Y_i - EY)) = 0, \quad i = 1, 2, \dots, n.$$

Hence, based on the estimator  $\hat{\psi}(u)$  in (7) of distorting function  $\psi(u)$ , we let

$$\hat{\tilde{Y}}_i = \hat{\psi}(U_i)\tilde{Y}, \quad i = 1, 2, \dots, n,$$

which can be taken as the approximations of actual responses. Denote  $\hat{\tilde{\mathbf{Y}}} = (\hat{\tilde{Y}}_1, \hat{\tilde{Y}}_2, \dots, \hat{\tilde{Y}}_n)^T$  and  $\tilde{\mathbf{Y}} = (\tilde{Y}_1, \tilde{Y}_2, \dots, \tilde{Y}_n)^T$ . The GCV criterion bases on the difference between  $\hat{\tilde{\mathbf{Y}}}$  and  $\tilde{\mathbf{Y}}$ . From (6) and (7), we have  $\hat{\tilde{\mathbf{Y}}} = \mathbf{W}\tilde{\mathbf{Y}}$ , where

$$\mathbf{W} = (W_{ij})_{n \times n}, \quad W_{ij} = \frac{K\left(\frac{U_i - U_j}{h_0}\right)}{\sum_{j=1}^n K\left(\frac{U_i - U_j}{h_0}\right)}, \quad i, j = 1, 2, \dots, n.$$

Thus, for the choice of the bandwidth  $h_0$ , we define

$$\text{GCV}(h_0) = \arg \min_{h_0} \left\{ \frac{n^{-1} \text{RSS}(h_0)}{[1 - \text{tr}(\mathbf{W})/n]^2} \right\},$$

where  $\text{tr}(\mathbf{W})$  is the trace of matrix  $\mathbf{W}$  and  $\text{RSS}(h_0) = \|\hat{\mathbf{Y}} - \tilde{\mathbf{Y}}\|^2 = \tilde{\mathbf{Y}}^T (\mathbf{I} - \mathbf{W})^2 \tilde{\mathbf{Y}}$  is the residual sum of squares. Then we minimize  $\text{GCV}(h_0)$  to obtain the bandwidth  $h_0$ . Similarly, the bandwidths  $h_j, j = 1, 2, \dots, p$  can be selected. Secondly, for the selection of two parameters  $\xi$  and  $\lambda$ , we minimize the BIC as (Li and Lu, 2018), which is given by

$$\text{BIC}(\xi, \lambda) = \log \left\{ \hat{\sigma}_{(\xi, \lambda)}^2 \right\} + \text{DF}_{(\xi, \lambda)} \log(n)/n,$$

where  $\hat{\sigma}_{(\xi, \lambda)}^2 = n^{-1} \|\hat{\mathbf{Y}} - \hat{\mathbf{X}}\hat{\boldsymbol{\beta}}^*\|^2$ , and  $\text{DF}_{(\xi, \lambda)}$  is the number of nonzero coefficients of  $\hat{\boldsymbol{\beta}}^*$ .

### 3 Simulation studies

In this section, simulation studies are conducted to assess the finite sample performance of our CBAR estimator, comparing it with some popular  $L_1$ -based variable selection methods. For example, the LASSO penalty  $p(|\beta_j|; \lambda_n) = \lambda_n |\beta_j|$ , the Adaptive LASSO (ALASSO) penalty  $p(|\beta_j|; \lambda_n) = \lambda_n \omega_j |\beta_j|$  with  $\omega_j$  being a weight, the SCAD penalty  $p(|\beta_j|; \lambda_n) = \lambda_n \int_0^{|\beta_j|} \min \{1, (a\lambda_n - x)/(a-1)\lambda_n\} dx$  with  $a > 2$  and the MCP penalty  $p(|\beta_j|; \lambda_n) = \int_0^{|\beta_j|} (\lambda_n - x/\tau_0)_+ dx$  with  $\tau_0 > 1$ . Furthermore, we include a method (denoted as NV-OLS) in which parameters are estimated by OLS with significant variables known in advance, to illustrate the performance of CBAR in more detail. As suggested by a reviewer, we add a method (denoted as LL-SCAD) in which local linear estimation (Delaigle et al., 2016) is first used to recover unobservable response and predictors, after which SCAD (Feng et al., 2020) is employed to select variables and estimate parameters simultaneously. The LASSO and ALASSO are implemented by using the R package *glmnet* and the SCAD and MCP are realized by the R package *ncvreg*. As noted by (Zhao et al., 2020; Li et al., 2021) and confirmed by our numerical studies (Supplementary Figure S.1), the CBAR estimator is not sensitive to the choice of  $\xi_n$ , thus we fix  $\xi_n = 100$  to reduce computational burden and use ten grid points evenly spaced on the interval  $[10^{-4}, 2 \log(p)]$  for  $\lambda_n$ . With respect to bandwidths  $h$ , ten grid points evenly spaced on the interval  $[1, 10]$  is used. We consider the following three case settings for model (1).

Case I: The predictor vector is generated from a  $p_n$ -dimension multivariate normality distribution, and the correlation coefficient between  $X_i$  and  $X_j$  is  $0.5^{|i-j|}$  for  $1 \leq i, j \leq p_n$ . The true parameter  $\boldsymbol{\beta}_{0n} = (4, 3, 0.5, 1.5, 1, 0, \dots, 0)^T$ , confounding variable  $U \sim N(5, 1)$ ,  $X_j$  and the distorting functions which satisfy the identifiability conditions are as follows

$$\begin{aligned}
X_1 &\sim N(3, 0.1), \phi_1(U) = \frac{(U+3)^2}{65}, \quad X_2 \sim N(2, 1), \phi_2(U) = \frac{(U+2)^2}{50}, \\
X_3 &\sim N(1.6, 0.3), \phi_3(U) = \frac{U+10}{15}, \quad X_4 \sim N(0.8, 0.5), \phi_4(U) = \frac{(U+1)^2}{37}, \\
X_5 &\sim N(1.2, 0.25), \phi_5(U) = \frac{U^2}{26}, \quad X_k \sim N(1, 0.2), \phi_k(U) = \frac{U-1}{4}, \quad k = 6, \dots, 20, \\
\psi(U) &= \frac{U+3}{8}.
\end{aligned}$$

We consider the model error  $\varepsilon$  follows  $N(0, 1)$  and  $N(0, 0.2^2)$ .

**Case II:** We consider there exist highly correlated variables. Let  $X_3 = X_2 + \varepsilon_0 Z_1$ ,  $X_1 = X_4 + \varepsilon_0 Z_2$ , where  $\varepsilon_0 = 0.3$ ,  $Z_i \sim N(0, 1)$ ,  $i = 1, 2$ . The correlation coefficients between  $X_3$  and  $X_2$ ,  $X_1$  and  $X_4$  are 0.958 and 0.921 respectively. The other setups are the same as in Case I.

**Case III:** To assess and compare the grouping effects of the five methods above and Elastic Net (Zou and Hastie, 2005), we conduct Case III. In model (1) we take  $n = 500$ ,  $p_n = 9$ ,  $\beta_{0n} = (2, 2, 2, 0.5, 0.5, 0.5, 0, 0, 0)^T$  and assume that

$$\begin{aligned}
X_i &= Z_3 + e_i, \quad Z_3 \sim N(0.15, 1), \quad \phi_i(U) = \frac{(U+3)^2}{65} && \text{for } i \in \{1, 2, 3\}, \\
X_i &= Z_4 + e_i, \quad Z_4 \sim N(0.15, 1), \quad \phi_i(U) = \frac{(U+2)^2}{50} && \text{for } i \in \{4, 5, 6\}, \\
X_i &\sim N(0.15, 1), \quad \phi_i(U) = \frac{(U+1)}{6} && \text{for } i \in \{7, 8, 9\}, \\
\psi(U) &= \frac{(U+3)}{8},
\end{aligned}$$

where  $\varepsilon \sim N(0, 0.2^2)$ ,  $e_i$  are independent  $N(0, 0.1^2)$  variables and  $U \sim N(5, 1)$ . In this setting,  $X_1$ ,  $X_2$  and  $X_3$  from a group with latent factor  $Z_3$ , and  $X_4$ ,  $X_5$  and  $X_6$  from another group with latent factor  $Z_4$ . The within group correlations are almost 1. Further, in the implementation of recovering the unobservable latent variable  $X = (X_1, X_2, \dots, X_p)^T$  and  $Y$ , the high-order kernel function  $K(t) = 15/32(3 - 7t^2)(1 - t^2)I(|t| \leq 1)$  is employed. The GCV and BIC are used to select bandwidths and tuning parameter, respectively. We report the empirical value of mean squared error (MSE), the averaged number of nonzero estimates of the parameters whose true values are zero (FP), the averaged number of zero estimates of the parameters whose true values are not zero (FN), the empirical probability that the selected model is exactly the true model (TM), a similarity measure (SM) between the selected set  $\hat{S}$  and the true active set  $S$ :  $\text{SM} = |\hat{S} \cap S|_0 / \sqrt{|\hat{S}|_0 |S|_0}$ , where  $|\cdot|_0$  denotes model size and Misclassification ( $\text{MisC} = \text{FN} + \text{FP}$ ). The true  $\sigma^2$  is estimated by  $(\hat{Y} - \hat{X}\hat{\beta})^T(\hat{Y} - \hat{X}\hat{\beta})/(n - \hat{q})$ , where  $\hat{q}$  is the selected non-zero number. We have run extensive simulations for a variety of settings by varying  $n$ ,  $p$  and  $\varepsilon$ , with 1000 replications for each setting. Part of the findings are presented in Table 1-Table 9. The additional results are given in Supplementary Material.

For Case I, one can see from Table 1-Table 4 that CBAR generally yields the largest TM and SM, and the smallest FP, MisC and MSE. For estimation of  $\beta$ , CBAR is superior to other methods, particularly in estimating weak signal ( $\beta_3 = 0.5$ ) when  $n$  or  $p$  is relatively large. For estimation of  $\sigma^2$ , no method dominates others across all scenarios and their performance are similar. For Case II, when there exist highly correlated variables, it is seen from Table 5-Table 8, CBAR still gives largest TM and SM, and the smallest FP, MisC. However, CBAR is slightly worse than LASSO in MSE but the difference gradually diminish as sample size increase. The instability and poor performance of LL-SCAD, particularly for the MSE and  $\hat{\sigma}^2$  in some scenarios, is due to the inaccuracy of the estimated distorted functions obtained via local linear estimation. Furthermore, we change the bandwidth value and observe from the results, which suggests that LL-SCAD is more sensitive to the selection of bandwidth than the SCAD and CBAR methods in the cases considered. Furthermore, unlike other methods, CBAR yields reliable estimates for both  $\beta$  and  $\sigma^2$  simultaneously. For Case I and II, Figure 1 and 2 present the TM and MSE results across various  $p_n$ , respectively. Because the MSE of LL-SCAD for case II is much higher than that of the other methods, its inclusion in Figure 1 and 2 would make it difficult to discern the differences among the remaining methods. Consequently, we have excluded the results of LL-SCAD. The results suggest that CBAR outperforms other methods, yielding the largest TM and the smallest or comparable MSE across all  $p_n$  considered.

Table 9 gives the grouping effects results for Case III. It can be seen that both CBAR and Elastic Net selected variables in two groups with almost same coefficients within each group, whereas LASSO, Adaptive LASSO, SCAD, and MCP failed to reveal any grouping information. Because the CBAR estimator is iteratively computed until the stop condition of algorithm is reached, we conduct additional simulations to record the times of iteration, which is plotted in Figure 3. Consider two scenarios (a). Case I:  $n = 100, p_n = 20, \varepsilon \sim N(0, 1), U \sim N(5, 1)$ ; (b). Case II:  $n = 100, p_n = 20, \varepsilon_0 = 0.3, \varepsilon \sim N(0, 1), U \sim N(5, 1)$ . It can be seen that after only 6 or 7 times of iteration, the estimates of zero coefficients shrink to 0 for the scenarios (a) and (b), respectively. For estimation of non-zero ones, the times of iteration needed are 11 and 15 for the scenarios (a) and (b), respectively. In conclusion, CBAR is a powerful and effective procedure in simultaneous variable selection and estimation and has a desirable advantage of more accurately selecting and estimating highly correlated variables.

## 4 Application to Boston Housing Data

In this section, we illustrate the method via an application to the Boston Housing data which has been intensively studied (see, e.g., (Li and Lu, 2018; Delaigle et al., 2016; Li et al., 2021)). The dataset comes from 506 census tracts in the Boston Standard Metropolitan Statistical Area in 1970. We are interested in investigating the relationships between the house price and the other various factors. The dataset includes 14-dimensional variables which may affect house price from different aspects such as location, size, social and natural environments. The explanation of these variables are

**Table 1** Variable selection and estimation results (with standard deviation in parentheses) for all methods under Case I with  $\varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$  across different  $(n, p)$  combinations

| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | MisC                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 100 | 20  | CBAR    | <b>0.4656</b> (0.4986) | <b>0.9337</b> (0.0701) | <b>0.0790</b> (0.3014) | 0.5540(0.5911)         | <b>0.6330</b> (0.6699) | <b>0.0450</b> (0.1372) |
|     |     | LASSO   | 0.3190(0.4663)         | NaN(NA)                | 1.2980(1.4210)         | <b>0.1060</b> (0.3506) | 1.4040(1.4411)         | 0.0505(0.0692)         |
|     |     | ALASSO  | 0.2050(0.4039)         | 0.8788(0.0889)         | 0.6600(1.0351)         | 0.6430(0.6180)         | 1.3030(1.0753)         | 0.0534(0.0838)         |
|     |     | SCAD    | 0.0900(0.2863)         | NaN(NA)                | 1.0350(1.4717)         | 0.8120(0.7178)         | 1.8470(1.3503)         | 0.0601(0.0788)         |
|     |     | LL-SCAD | 0.0970(0.2961)         | 0.8304(0.0949)         | 1.0660(1.4737)         | 0.8350(0.7445)         | 1.9010(1.3360)         | 0.1053(0.2336)         |
|     |     | MCP     | 0.1400(0.3472)         | 0.8599(0.0922)         | 0.7970(1.1913)         | 0.7320(0.6266)         | 1.5290(1.2002)         | 0.0614(0.2153)         |
| 50  | 50  | CBAR    | <b>0.4460</b> (0.4973) | <b>0.9287</b> (0.0751) | <b>0.0770</b> (0.3274) | 0.6010(0.6342)         | <b>0.6780</b> (0.7161) | <b>0.0174</b> (0.0271) |
|     |     | LASSO   | 0.2790(0.4487)         | 0.8760(0.1068)         | 1.5910(1.8340)         | <b>0.1370</b> (0.3666) | 1.7280(1.8427)         | 0.0223(0.0218)         |
|     |     | ALASSO  | 0.1070(0.3093)         | 0.8283(0.1101)         | 1.1970(1.8525)         | 0.8190(0.6683)         | 2.0160(1.8042)         | 0.0250(0.0255)         |
|     |     | SCAD    | 0.0360(0.1864)         | 0.7913(0.1040)         | 1.5050(2.5762)         | 1.0550(0.7173)         | 2.5600(2.3345)         | 0.0289(0.0332)         |
|     |     | LL-SCAD | 0.0380(0.1913)         | 0.7824(0.1132)         | 1.7370(2.9904)         | 1.0550(0.7256)         | 2.7920(2.7532)         | 0.0498(0.1143)         |
|     |     | MCP     | 0.0400(0.1961)         | 0.7920(0.1184)         | 1.8900(2.7531)         | 0.8640(0.6663)         | 2.7540(2.5733)         | 0.0290(0.0343)         |
| 80  | 80  | CBAR    | <b>0.4030</b> (0.4907) | <b>0.9232</b> (0.0742) | <b>0.0850</b> (0.3375) | 0.6460(0.6333)         | <b>0.7310</b> (0.7037) | <b>0.0113</b> (0.0187) |
|     |     | LASSO   | 0.2570(0.4372)         | 0.8704(0.1107)         | 1.7600(2.5284)         | <b>0.1320</b> (0.3588) | 1.8920(2.5244)         | 0.0153(0.0148)         |
|     |     | ALASSO  | 0.0460(0.2096)         | 0.7733(0.1205)         | 1.7420(3.3046)         | 1.0730(0.6857)         | 2.8150(3.2204)         | 0.0216(0.0208)         |
|     |     | SCAD    | 0.0270(0.1622)         | 0.7661(0.1388)         | 2.8030(6.7928)         | 1.0930(0.6759)         | 3.8960(6.5875)         | 0.0272(0.0471)         |
|     |     | LL-SCAD | 0.0170(0.1293)         | 0.7404(0.1539)         | 3.7490(7.9145)         | 1.1060(0.7136)         | 4.8550(7.6546)         | 0.0492(0.1000)         |
|     |     | MCP     | 0.0240(0.1531)         | 0.7114(0.1738)         | 5.0510(7.4163)         | 0.8580(0.6498)         | 5.9090(7.1971)         | 0.0363(0.0556)         |
| 200 | 20  | CBAR    | <b>0.7230</b> (0.4477) | <b>0.9709</b> (0.0474) | <b>0.0270</b> (0.1622) | 0.2530(0.4349)         | <b>0.2800</b> (0.4559) | <b>0.0160</b> (0.0250) |
|     |     | LASSO   | 0.4630(0.4989)         | 0.9294(0.0790)         | 0.9120(1.1626)         | <b>0.0130</b> (0.1133) | 0.9250(1.1648)         | 0.0197(0.0249)         |
|     |     | ALASSO  | 0.4520(0.4979)         | 0.9353(0.0670)         | 0.3890(0.7565)         | 0.3160(0.4694)         | 0.7050(0.8016)         | 0.0197(0.0250)         |
|     |     | SCAD    | 0.2460(0.4309)         | 0.9020(0.0731)         | 0.6630(1.0481)         | 0.4410(0.5184)         | 1.1040(0.9890)         | 0.0229(0.0279)         |
|     |     | LL-SCAD | 0.2400(0.4273)         | 0.8902(0.0854)         | 0.8290(1.2366)         | 0.4450(0.5756)         | 1.2740(1.1935)         | 0.0599(0.1801)         |
|     |     | MCP     | 0.3670(0.4822)         | 0.9240(0.0678)         | 0.4170(0.7998)         | 0.4030(0.4968)         | 0.8200(0.8236)         | 0.0208(0.0264)         |
| 50  | 50  | CBAR    | <b>0.7200</b> (0.4492) | <b>0.9699</b> (0.0494) | <b>0.0240</b> (0.1595) | 0.2650(0.4461)         | <b>0.2890</b> (0.4751) | <b>0.0061</b> (0.0093) |
|     |     | LASSO   | 0.4300(0.4953)         | 0.9181(0.0890)         | 1.0880(1.3814)         | <b>0.0220</b> (0.1468) | 1.1100(1.3942)         | 0.0089(0.0088)         |
|     |     | ALASSO  | 0.3180(0.4659)         | 0.9068(0.0829)         | 0.6680(1.1692)         | 0.4090(0.5020)         | 1.0770(1.1612)         | 0.0088(0.0098)         |
|     |     | SCAD    | 0.1330(0.3397)         | 0.8635(0.0841)         | 1.0010(1.4265)         | 0.5900(0.5497)         | 1.5910(1.3090)         | 0.0108(0.0110)         |

Table 1 (continued)

| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | MisC                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 400 | 80  | LL-SCAD | 0.1440(0.3513)         | 0.8556(0.0933)         | 1.1410(1.5993)         | 0.5800(0.6050)         | 1.7210(1.5051)         | 0.0199(0.0462)         |
|     |     | MCP     | 0.2130(0.4096)         | 0.8867(0.0842)         | 0.8040(1.3332)         | 0.5120(0.5217)         | 1.3160(1.2751)         | 0.0092(0.0106)         |
|     |     | CBAR    | <b>0.7650</b> (0.4242) | <b>0.9748</b> (0.0461) | <b>0.0110</b> (0.1375) | 0.2300(0.4258)         | <b>0.2410</b> (0.4440) | <b>0.0035</b> (0.0046) |
|     |     | LASSO   | 0.4230(0.4943)         | 0.9171(0.0878)         | 1.1190(1.5580)         | <b>0.0160</b> (0.1255) | 1.1350(1.5586)         | 0.0059(0.0048)         |
|     |     | ALASSO  | 0.2800(0.4492)         | 0.8991(0.0855)         | 0.7270(1.3184)         | 0.4500(0.5136)         | 1.1770(1.2905)         | 0.0058(0.0052)         |
|     |     | SCAD    | 0.1200(0.3251)         | 0.8529(0.0900)         | 1.1040(1.6005)         | 0.6340(0.5660)         | 1.7380(1.4824)         | 0.0074(0.0060)         |
|     |     | LL-SCAD | 0.1050(0.3067)         | 0.8339(0.1017)         | 1.4290(2.0869)         | 0.6600(0.6394)         | 2.0890(1.9300)         | 0.0132(0.0255)         |
|     |     | MCP     | 0.2130(0.4096)         | 0.8867(0.0842)         | 0.8040(1.3332)         | 0.5120(0.5217)         | 1.3160(1.2751)         | 0.0057(0.0067)         |
|     |     | CBAR    | <b>0.9370</b> (0.2431) | <b>0.9935</b> (0.0251) | <b>0.0090</b> (0.0945) | 0.0540(0.2261)         | <b>0.0630</b> (0.2431) | <b>0.0050</b> (0.0052) |
|     |     | LASSO   | 0.5270(0.4995)         | 0.9423(0.0698)         | 0.7350(0.9652)         | <b>0</b> (0)           | 0.7350(0.9652)         | 0.0076(0.0057)         |
| 50  | 20  | ALASSO  | 0.7390(0.4394)         | 0.9729(0.0477)         | 0.2220(0.5089)         | 0.0810(0.2730)         | 0.3030(0.5509)         | 0.0067(0.0060)         |
|     |     | SCAD    | 0.5040(0.5002)         | 0.9465(0.0588)         | 0.4210(0.7143)         | 0.1820(0.3860)         | 0.6030(0.7183)         | 0.0081(0.0068)         |
|     |     | LL-SCAD | 0.5050(0.5002)         | 0.9426(0.0663)         | 0.5080(0.8653)         | 0.1630(0.3829)         | 0.6710(0.8646)         | 0.0283(0.1171)         |
|     |     | MCP     | 0.6770(0.4679)         | 0.9663(0.0514)         | 0.2650(0.5595)         | 0.1110(0.3143)         | 0.3760(0.5992)         | 0.0066(0.0060)         |
|     |     | CBAR    | <b>0.9600</b> (0.1961) | <b>0.9959</b> (0.0203) | <b>0.0050</b> (0.0706) | 0.0350(0.1839)         | <b>0.0400</b> (0.1961) | <b>0.0019</b> (0.0019) |
|     |     | LASSO   | 0.4630(0.4989)         | 0.9294(0.0790)         | 0.9120(1.1626)         | <b>0.0130</b> (0.1133) | 0.9250(1.1648)         | 0.0079(0.0100)         |
|     |     | ALASSO  | 0.6450(0.4788)         | 0.9579(0.0631)         | 0.3950(0.8341)         | 0.1070(0.3093)         | 0.5020(0.8478)         | 0.0031(0.0025)         |
|     |     | SCAD    | 0.3620(0.4808)         | 0.9220(0.0705)         | 0.6120(1.0680)         | 0.2990(0.4580)         | 0.9110(1.0120)         | 0.0043(0.0031)         |
|     |     | LL-SCAD | 0.3220(0.4675)         | 0.9084(0.0816)         | 0.8390(1.2307)         | 0.2730(0.4655)         | 1.1120(1.1954)         | 0.0130(0.0543)         |
|     |     | MCP     | 0.5160(0.5000)         | 0.9425(0.0672)         | 0.4990(0.9246)         | 0.1790(0.3835)         | 0.6780(0.9150)         | 0.0032(0.0027)         |
| 80  | 20  | CBAR    | <b>0.9380</b> (0.2413) | <b>0.9934</b> (0.0259) | <b>0.0030</b> (0.0547) | 0.0600(0.2376)         | <b>0.0630</b> (0.2472) | <b>0.0013</b> (0.0017) |
|     |     | LASSO   | 0.5240(0.4997)         | 0.9395(0.0734)         | 0.7830(1.0474)         | <b>0</b> (0)           | 0.7830(1.0474)         | 0.0026(0.0020)         |
|     |     | ALASSO  | 0.5840(0.4931)         | 0.9518(0.0632)         | 0.4000(0.7711)         | 0.1540(0.3611)         | 0.5540(0.7961)         | 0.0022(0.0020)         |
|     |     | SCAD    | 0.3360(0.4726)         | 0.9170(0.0709)         | 0.5970(1.0127)         | 0.3510(0.4775)         | 0.9480(0.9634)         | 0.0032(0.0023)         |
|     |     | LL-SCAD | 0.2670(0.4426)         | 0.9007(0.0805)         | 0.8610(1.3826)         | 0.3370(0.4896)         | 1.1980(1.3201)         | 0.0073(0.0301)         |
|     |     | MCP     | 0.4560(0.4983)         | 0.9330(0.0716)         | 0.5300(1.0120)         | 0.2530(0.4349)         | 0.7830(0.9964)         | 0.0024(0.0021)         |

Note 1. Boldface indicates the best performance among the five methods.

Note 2. The notation "NaN" indicates cases where the SM metric is undefined. This occurs when  $|\hat{S}|_0$  is zero in the denominator

**Table 2** Variable selection and estimation results (with standard deviation in parentheses) for all methods under Case 1 with  $\varepsilon \sim N(0, 1)$ ,  $U \sim N(5, 1)$  and fixed  $n = 400$ 

| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | Misc                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 400 | 100 | CBAR    | <b>0.9650</b> (0.1839) | <b>0.9962</b> (0.0206) | <b>0.0040</b> (0.0632) | 0.0330(0.1787)         | <b>0.0370</b> (0.1992) | <b>0.0010</b> (0.0013) |
|     |     | LASSO   | 0.5020(0.5002)         | 0.9339(0.0792)         | 0.8750(1.1894)         | <b>0.0020</b> (0.0447) | 0.8770(1.1905)         | 0.0021(0.0016)         |
|     |     | ALASSO  | 0.5860(0.4928)         | 0.9503(0.0669)         | 0.4440(0.8518)         | 0.1410(0.3482)         | 0.5850(0.8750)         | 0.0018(0.0017)         |
|     |     | SCAD    | 0.3070(0.4615)         | 0.9115(0.0730)         | 0.6880(1.0751)         | 0.3410(0.4764)         | 1.0290(1.0095)         | 0.0026(0.0020)         |
|     |     | LL-SCAD | 0.2950(0.4563)         | 0.9001(0.0849)         | 0.9320(1.4019)         | 0.3040(0.4710)         | 1.2360(1.3334)         | 0.0062(0.0236)         |
|     |     | MCP     | 0.4230(0.4943)         | 0.9279(0.0744)         | 0.6140(1.0919)         | 0.2450(0.4326)         | 0.8590(1.0688)         | 0.0019(0.0018)         |
| 200 | 200 | CBAR    | <b>0.9490</b> (0.2201) | <b>0.9946</b> (0.0237) | <b>0.0060</b> (0.1094) | 0.0470(0.2117)         | <b>0.0530</b> (0.2372) | <b>0.0005</b> (0.0008) |
|     |     | LASSO   | 0.5360(0.4990)         | 0.9387(0.0781)         | 0.8100(1.1753)         | <b>0.0030</b> (0.0547) | 0.8130(1.1753)         | 0.0012(0.0010)         |
|     |     | ALASSO  | 0.4690(0.4993)         | 0.9312(0.0775)         | 0.5740(1.0486)         | 0.2390(0.4267)         | 0.8130(1.0636)         | 0.0011(0.0010)         |
|     |     | SCAD    | 0.2600(0.4389)         | 0.9008(0.0769)         | 0.7440(1.1823)         | 0.4070(0.4915)         | 1.1510(1.1169)         | 0.0015(0.0011)         |
|     |     | LL-SCAD | 0.2330(0.4230)         | 0.8805(0.0941)         | 1.1740(1.7084)         | 0.3580(0.4900)         | 1.5320(1.6241)         | 0.0032(0.0106)         |
|     |     | MCP     | 0.3310(0.4708)         | 0.9135(0.0754)         | 0.6680(1.1503)         | 0.3440(0.4753)         | 1.0120(1.0968)         | 0.0011(0.0010)         |

**Table 3** [Case I :  $p_n = 20, \varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$ ] Parameter estimation and standard deviation (in parentheses) for all methods

| $(p_n, n)$ | Method  | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | $\hat{\beta}_4$ | $\hat{\beta}_5$ | $\hat{\sigma}^2$ |
|------------|---------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
| (20,100)   | CBAR    | 3.8641(0.5062)  | 3.0238(0.2475)  | 0.4052(0.4264)  | 1.4647(0.3210)  | 0.9122(0.3913)  | 1.3002(1.8026)   |
|            | LASSO   | 3.5347(0.5164)  | 2.9273(0.2422)  | 0.4779(0.3071)  | 1.3178(0.2879)  | 0.7612(0.3067)  | 1.3567(2.5527)   |
|            | ALASSO  | 3.7478(0.5399)  | 3.0644(0.2529)  | 0.2932(0.3785)  | 1.4870(0.3315)  | 0.7753(0.4025)  | 1.3095(2.2673)   |
|            | SCAD    | 3.8830(0.5262)  | 3.0564(0.2519)  | 0.2601(0.4016)  | 1.5735(0.3618)  | 0.7430(0.4906)  | 1.3297(2.5517)   |
|            | LL-SCAD | 4.0420(0.8248)  | 2.9638(0.4681)  | 0.3468(0.5428)  | 1.5242(0.5035)  | 0.7303(0.5646)  | 2.2154(19.9345)  |
| (20,200)   | MCP     | 3.8940(0.5041)  | 3.0428(0.2430)  | 0.3175(0.4455)  | 1.5227(0.3430)  | 0.8513(0.4514)  | 1.2784(1.7186)   |
|            | NV-OLS  | 3.8929(0.5095)  | 2.9842(0.2395)  | 0.5780(0.3331)  | 1.4002(0.2877)  | 0.9974(0.3081)  | 1.3113(2.1290)   |
|            | CBAR    | 3.9597(0.3240)  | 3.0222(0.1380)  | 0.4435(0.2933)  | 1.4752(0.2242)  | 0.9947(0.2156)  | 1.1190(0.1985)   |
|            | LASSO   | 3.7224(0.3291)  | 2.9606(0.1309)  | 0.4662(0.2004)  | 1.3896(0.2074)  | 0.8485(0.2076)  | 1.1406(0.2078)   |
|            | ALASSO  | 3.9095(0.3332)  | 3.0527(0.1376)  | 0.3343(0.2733)  | 1.5104(0.2312)  | 0.9144(0.2386)  | 1.1191(0.2014)   |
| (20,400)   | SCAD    | 3.9734(0.3283)  | 3.0547(0.1400)  | 0.2913(0.3196)  | 1.5441(0.2489)  | 0.9450(0.2827)  | 1.1258(0.2076)   |
|            | LL-SCAD | 4.0521(0.6141)  | 2.9673(0.4326)  | 0.3991(0.4780)  | 1.5221(0.3694)  | 0.8870(0.3898)  | 1.4276(1.0349)   |
|            | MCP     | 3.9720(0.3247)  | 3.0418(0.1400)  | 0.3487(0.3248)  | 1.5095(0.2377)  | 0.9948(0.2396)  | 1.1170(0.2055)   |
|            | NV-OLS  | 3.9722(0.3212)  | 2.9998(0.1275)  | 0.5379(0.2034)  | 1.4456(0.2065)  | 1.0163(0.2025)  | 1.1170(0.1986)   |
|            | CBAR    | 3.9885(0.1941)  | 3.0192(0.0784)  | 0.4854(0.1624)  | 1.5046(0.1112)  | 0.9928(0.1279)  | 1.0581(0.0992)   |
| (20,600)   | LASSO   | 3.8146(0.1975)  | 2.9843(0.0743)  | 0.4637(0.1265)  | 1.4555(0.1072)  | 0.8822(0.1294)  | 1.0708(0.1009)   |
|            | ALASSO  | 3.9607(0.1960)  | 3.0422(0.0796)  | 0.3971(0.1731)  | 1.5338(0.1150)  | 0.9497(0.1365)  | 1.0587(0.0996)   |
|            | SCAD    | 3.9927(0.1950)  | 3.0422(0.0860)  | 0.3804(0.2372)  | 1.5387(0.1240)  | 0.9948(0.1364)  | 1.0613(0.1003)   |
|            | LL-SCAD | 4.0064(0.4543)  | 2.9930(0.3020)  | 0.4657(0.3432)  | 1.4989(0.2536)  | 0.9739(0.2230)  | 1.3929(5.2186)   |
|            | MCP     | 3.9931(0.1943)  | 3.0267(0.0826)  | 0.4492(0.2062)  | 1.5156(0.1173)  | 1.0006(0.1276)  | 1.0568(0.0996)   |
| (20,600)   | NV-OLS  | 3.9933(0.1935)  | 3.0120(0.0737)  | 0.5153(0.1254)  | 1.4949(0.1061)  | 1.0021(0.1258)  | 1.0577(0.0993)   |
|            | CBAR    | 3.9955(0.1559)  | 3.0163(0.0617)  | 0.4975(0.1039)  | 1.5059(0.0809)  | 0.9932(0.0973)  | 1.0383(0.0757)   |
|            | LASSO   | 3.8486(0.1610)  | 2.9898(0.0613)  | 0.4659(0.0992)  | 1.4688(0.0809)  | 0.8993(0.1013)  | 1.0480(0.0769)   |
|            | ALASSO  | 3.9756(0.1569)  | 3.0342(0.0635)  | 0.4270(0.1191)  | 1.5290(0.0840)  | 0.9622(0.1015)  | 1.0386(0.0758)   |
|            | SCAD    | 3.9981(0.1558)  | 3.0283(0.0676)  | 0.4423(0.1672)  | 1.5228(0.0903)  | 0.9987(0.0974)  | 1.0389(0.0761)   |
| (20,600)   | LL-SCAD | 3.9959(0.3731)  | 2.9977(0.2673)  | 0.4961(0.2737)  | 1.4972(0.2268)  | 0.9750(0.1807)  | 1.2124(1.1209)   |
|            | MCP     | 3.9982(0.1557)  | 3.0180(0.0641)  | 0.4889(0.1308)  | 1.5082(0.0836)  | 0.9986(0.0969)  | 1.0372(0.0757)   |
|            | NV-OLS  | 3.9986(0.1558)  | 3.0133(0.0612)  | 0.5096(0.0981)  | 1.5019(0.0799)  | 0.9990(0.0965)  | 1.0383(0.0758)   |

summarized in Supplementary Table S.10. The basic equation (Harrison and Rubinfeld, 1978) employed is hedonic housing value equation, which is defined as

$$\begin{aligned} \log(\text{MV}) = & \beta_0 + \beta_1 \text{RM}^2 + \beta_2 \text{AGE} + \beta_3 \log(\text{DIS}) + \beta_4 \log(\text{RAD}) + \beta_5 \text{TAX} \\ & + \beta_6 \text{PTRATIO} + \beta_7 (\text{B} - 0.63)^2 + \beta_8 \log(\text{LSTAT}) + \beta_9 \text{CRIM} \quad (14) \\ & + \beta_{10} \text{ZN} + \beta_{11} \text{INDUS} + \beta_{12} \text{CHAS} + \beta_{13} \text{NOX}^p + \varepsilon, \end{aligned}$$

where the log transformation of variables MV, DIS, RAD and LSTAT and power transformation of RM and NOX were proved to fit the Boston Housing data well. The best value of exponent  $p$  for NOX was estimated is 2 by performing a grid search. Zhang et al. (2012a) adopted pupil-teacher ratio by town (PTRATIO) as  $U$  and found that the selection of such confounding variables was more reasonable, and PTRATIO had impact on other variables, thus truly revealing significant predictors in the model. To make the relationships more apparent, we compute the correlation coefficients between PTRATIO and MV as well as between PTRATIO and each of the other predictors. Subsequently, considering PTRATIO as a confounding variable, we obtain estimates of the real response and predictors, and then calculate the correlation coefficients between PTRATIO and the recovered response MV and the predictors. The results, presented in Table 10, show that the correlations are substantially reduced after the calibration. This suggests that the recovered data are approximately independent of PTRATIO. Notice that while Table 10 reveals the near-zero correlations after recovery, Assumption (b) requires the mutual independence between confounding variable  $U$  and the real variables  $X$  and  $Y$ , thus this assumption may be approximately satisfied in Boston Housing data rather than exactly satisfied. Furthermore, we present the patterns of  $\psi(u)$  and  $\phi_j(u)$ ,  $j = 1, 2, \dots, 12$  against PTRATIO in Supplementary Figure S.2 and S.3, respectively. These plots indicate that  $\psi(u)$  and  $\phi_j(u)$ ,  $j = 1, 2, \dots, 12$  are not linear, suggesting the confounding variable PTRATIO indeed has impact on house price (MV) and the other predictors. So we let PTRATIO as  $U$ .

The estimated coefficients result in model (14) is reported in Table 11. One can see from Table 11: (1) The variables selected by CBAR are also identified by other methods, with the exception of MCP, which fails to identify RM as significant. This result seems unreasonable because generally, the larger the house (with more rooms), the higher its price. Overall, the CBAR method yields the sparsest model with 5 predictors, which is consistent with our findings in simulation studies and achieves the smallest BIC value among all competing methods. (2) The black proportion of population (B) is an unimportant factor for the house price selected by CBAR method, which is in line with the actual situation. But there is a positive correlation between the house price and B (house price increases when B increases) in (Li and Lu, 2018), which does not reflect the reality. (3) Compared with Table 2 in (Li et al., 2014), the main differences are that NOX will affect the house price in this paper while not in (Li et al., 2014), and AGE is an unimportant factor for the house value here which means that there is no significant discrepancy between the earlier and recent houses. For comparison, we also present the results without considering confounding variable, where PTRATIO is included as a predictor in Supplementary Table S.11.

**Table 4** [Case I :  $n = 400$ ,  $\varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$ ] Parameter estimation and standard deviation (in parentheses) for all methods

| $(p_n, n)$ | Method  | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | $\hat{\beta}_4$ | $\hat{\beta}_5$ | $\hat{\sigma}^2$ |
|------------|---------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
| (50,400)   | CBAR    | 3.9913(0.1941)  | 3.0215(0.0769)  | 0.4858(0.1525)  | 1.5030(0.1082)  | 0.9916(0.1227)  | 1.1915(0.1106)   |
|            | LASSO   | 3.7634(0.2013)  | 2.9776(0.0756)  | 0.4491(0.1258)  | 1.4408(0.1048)  | 0.8494(0.1223)  | 1.0794(0.1020)   |
|            | ALASSO  | 3.9524(0.1978)  | 3.0555(0.0799)  | 0.3535(0.1776)  | 1.5466(0.1173)  | 0.9294(0.1376)  | 0.9899(0.0169)   |
|            | SCAD    | 3.9958(0.1958)  | 3.0701(0.0871)  | 0.2637(0.2401)  | 1.5709(0.1297)  | 0.9813(0.1456)  | 0.9817(0.0202)   |
|            | LL-SCAD | 4.0168(0.4250)  | 3.0194(0.3050)  | 0.3482(0.3621)  | 1.5511(0.2681)  | 0.9397(0.2589)  | 1.2787(1.0847)   |
|            | MCP     | 3.9928(0.1907)  | 3.0431(0.0888)  | 0.3757(0.2360)  | 1.5319(0.1290)  | 1.0049(0.1308)  | 0.9864(0.0183)   |
| (80,400)   | NV-OLS  | 3.9975(0.1934)  | 3.0140(0.0746)  | 0.5163(0.1258)  | 1.4929(0.1036)  | 1.0019(0.1211)  | 1.0554(0.0979)   |
|            | CBAR    | 3.9859(0.1954)  | 3.0222(0.0818)  | 0.4737(0.1691)  | 1.4992(0.1187)  | 0.9996(0.1319)  | 0.9992(0.0031)   |
|            | LASSO   | 3.7322(0.1993)  | 2.9719(0.0789)  | 0.4378(0.1324)  | 1.4278(0.1139)  | 0.8438(0.1327)  | 0.9902(0.0131)   |
|            | ALASSO  | 3.9397(0.2009)  | 3.0614(0.0851)  | 0.3204(0.1902)  | 1.5491(0.1276)  | 0.9260(0.1480)  | 0.9930(0.0099)   |
|            | SCAD    | 3.9924(0.1981)  | 3.0800(0.0889)  | 0.2081(0.2231)  | 1.5960(0.1407)  | 0.9671(0.1773)  | 0.9881(0.0120)   |
|            | LL-SCAD | 3.9926(0.3811)  | 3.0436(0.2586)  | 0.2699(0.3281)  | 1.5828(0.2554)  | 0.9176(0.2625)  | 1.2710(0.9590)   |
| (100,400)  | MCP     | 3.9924(0.1971)  | 3.0525(0.0907)  | 0.3324(0.2451)  | 1.5438(0.1330)  | 1.0067(0.1388)  | 0.9902(0.0124)   |
|            | NV-OLS  | 3.9927(0.1949)  | 3.0127(0.0781)  | 0.5129(0.1301)  | 1.4867(0.1125)  | 1.0110(0.1279)  | 1.0571(0.1060)   |
|            | CBAR    | 3.9938(0.1898)  | 3.0153(0.0778)  | 0.4964(0.1548)  | 1.4973(0.1237)  | 0.9973(0.1375)  | 1.0000(0.0019)   |
|            | LASSO   | 3.7344(0.1942)  | 2.9650(0.0759)  | 0.4538(0.1313)  | 1.4265(0.1214)  | 0.8381(0.1374)  | 0.9912(0.0119)   |
|            | ALASSO  | 3.9438(0.1940)  | 3.0582(0.0824)  | 0.3287(0.1902)  | 1.5522(0.1357)  | 0.9167(0.1552)  | 0.9941(0.0087)   |
|            | SCAD    | 4.0019(0.1915)  | 3.0774(0.0861)  | 0.2113(0.2251)  | 1.6007(0.1476)  | 0.9642(0.1834)  | 0.9897(0.0101)   |
| (100,400)  | LL-SCAD | 4.0173(0.3969)  | 3.0341(0.2871)  | 0.2884(0.3671)  | 1.5796(0.2777)  | 0.9209(0.2731)  | 1.2267(0.5353)   |
|            | MCP     | 4.0010(0.1899)  | 3.0487(0.0897)  | 0.3416(0.2482)  | 1.5462(0.1423)  | 1.0052(0.1452)  | 0.9914(0.0106)   |
|            | NV-OLS  | 4.0008(0.1893)  | 3.0071(0.0753)  | 0.5296(0.1290)  | 1.4867(0.1203)  | 1.0092(0.1336)  | 1.0552(0.1006)   |

**Table 5** Variable selection and estimation results (with standard deviation in parentheses) for all methods under Case II with  $\varepsilon_0 = 0.3, \varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$ .

| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | MisC                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 100 | 20  | CBAR    | <b>0.3730</b> (0.4838) | <b>0.9104</b> (0.0877) | <b>0.1040</b> (0.5792) | 0.7460(0.7266)         | <b>0.8500</b> (0.9101) | 0.1470(0.2614)         |
|     |     | LASSO   | 0.3630(0.4811)         | 0.9022(0.0948)         | 1.0810(1.3335)         | <b>0.1650</b> (0.4024) | 1.2460(1.3753)         | <b>0.1191</b> (0.2068) |
|     |     | ALASSO  | 0.1940(0.3956)         | 0.8608(0.1023)         | 0.6100(0.9438)         | 0.8170(0.7849)         | 1.4270(1.0825)         | 0.1587(0.2498)         |
|     |     | SCAD    | 0.0060(0.0773)         | 0.7332(0.0960)         | 1.6590(1.7349)         | 1.3170(0.8398)         | 2.9760(1.4414)         | 0.2266(0.3002)         |
|     |     | LL-SCAD | 0.0030(0.0547)         | NaN(NA)                | 1.4240(1.6861)         | 1.5720(0.8738)         | 2.9960(1.4691)         | 1.1784(14.8054)        |
|     | 50  | MCP     | 0.0150(0.1216)         | 0.7627(0.1002)         | 1.3820(1.5279)         | 1.2110(0.7753)         | 2.5930(1.3687)         | 0.2057(0.2641)         |
|     |     | CBAR    | <b>0.3610</b> (0.4682) | <b>0.9041</b> (0.0870) | <b>0.1230</b> (0.3899) | 0.7810(0.7082)         | <b>0.9040</b> (0.7996) | 0.0599(0.1074)         |
|     |     | LASSO   | 0.3601(0.4830)         | 0.8939(0.1081)         | 1.1810(1.7724)         | <b>0.2140</b> (0.4565) | 1.3950(1.8309)         | <b>0.0484</b> (0.0858) |
|     |     | ALASSO  | 0.0790(0.2699)         | 0.7960(0.1192)         | 1.1970(1.8220)         | 1.0730(0.8101)         | 2.2700(1.7917)         | 0.0773(0.1146)         |
|     |     | SCAD    | 0(0)                   | 0.6313(0.1320)         | 4.0230(7.8966)         | 1.8790(0.8069)         | 5.9020(7.4439)         | 0.0934(0.1068)         |
| 200 | 80  | LL-SCAD | 0.0020(0.0447)         | NaN(NA)                | 2.1310(3.1705)         | 1.8980(0.7799)         | 4.0290(2.8160)         | 0.1862(0.7349)         |
|     |     | MCP     | 0.0010(0.0316)         | 0.6669(0.1254)         | 2.8820(3.4808)         | 1.5250(0.7348)         | 4.4070(3.2024)         | 0.1072(0.1238)         |
|     |     | CBAR    | <b>0.3610</b> (0.4805) | <b>0.9099</b> (0.0855) | <b>0.1320</b> (0.4591) | 0.7270(0.6949)         | <b>0.8590</b> (0.8354) | 0.0372(0.0892)         |
|     |     | LASSO   | 0.3240(0.4682)         | 0.8811(0.1128)         | 1.4010(1.8427)         | <b>0.2110</b> (0.4568) | 1.6120(1.8839)         | <b>0.0299</b> (0.0493) |
|     |     | ALASSO  | 0.0410(0.1984)         | 0.7489(0.1241)         | 1.3550(1.9927)         | 1.3860(0.8943)         | 2.7410(1.9099)         | 0.0654(0.0814)         |
|     |     | SCAD    | 0(0)                   | 0.6313(0.132)          | 4.0230(7.8966)         | 1.8790(0.8069)         | 5.9020(7.4439)         | 0.0934(0.1068)         |
|     |     | LL-SCAD | 0(0)                   | NaN(NA)                | 4.7440(8.5692)         | 1.9410(0.8356)         | 6.6850(8.1719)         | 0.4603(5.2711)         |
|     |     | MCP     | 0.0010(0.0316)         | 0.5908(0.1686)         | 6.3490(7.6951)         | 1.5460(0.7446)         | 7.8950(7.4009)         | 0.1026(0.3262)         |
|     |     | CBAR    | <b>0.5160</b> (0.5000) | <b>0.9460</b> (0.0591) | <b>0.0330</b> (0.1948) | 0.4820(0.5310)         | <b>0.5150</b> (0.5622) | 0.0600(0.1356)         |
|     |     | LASSO   | 0.5040(0.5002)         | 0.9370(0.0748)         | 0.7360(1.0086)         | <b>0.0530</b> (0.2329) | 0.7890(1.0254)         | <b>0.0514</b> (0.1231) |
| 500 | 200 | ALASSO  | 0.3620(0.4808)         | 0.9190(0.0731)         | 0.3330(0.7118)         | 0.5060(0.548)          | 0.8390(0.8184)         | 0.0639(0.1357)         |
|     |     | SCAD    | 0.0200(0.1401)         | 0.8100(0.0851)         | 1.2100(1.4325)         | 0.9240(0.5390)         | 2.1340(1.2750)         | 0.0871(0.1519)         |
|     |     | LL-SCAD | 0.0150(0.1216)         | NaN(NA)                | 1.2660(1.4857)         | 1.0330(0.6499)         | 2.2990(1.3503)         | 1.0920(27.0707)        |
|     |     | MCP     | 0.0380(0.1913)         | 0.8377(0.0800)         | 0.8940(1.2009)         | 0.8690(0.5255)         | 1.7630(1.0798)         | 0.0808(0.1471)         |
|     |     | CBAR    | <b>0.5120</b> (0.5001) | <b>0.9462</b> (0.0581) | <b>0.0340</b> (0.2165) | 0.4800(0.5271)         | <b>0.5140</b> (0.5569) | 0.0226(0.0513)         |
|     | 50  | LASSO   | 0.4410(0.4968)         | 0.9217(0.0858)         | 0.9730(1.3000)         | <b>0.0570</b> (0.2320) | 1.0300(1.3095)         | <b>0.0194</b> (0.0458) |
|     |     | ALASSO  | 0.2520(0.4344)         | 0.8938(0.0846)         | 0.5890(1.0291)         | 0.5650(0.5640)         | 1.1540(1.0693)         | 0.0270(0.0583)         |
|     |     | SCAD    | 0(0)                   | 0.7297(0.0966)         | 2.0730(2.3459)         | 1.2270(0.5864)         | 3.3000(2.0519)         | 0.0496(0.0719)         |

Table 5 (continued)

| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | MisC                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 80  |     | LL-SCAD | 0.0050(0.0706)         | NaN(NA)                | 2.1340(2.4556)         | 1.3270(0.6860)         | 3.4610(2.1980)         | 0.1357(1.0683)         |
|     |     | MCP     | 0.0020(0.0447)         | 0.7570(0.1037)         | 1.9440(2.3234)         | 1.0730(0.5215)         | 3.0170(2.1031)         | 0.0400(0.0632)         |
|     |     | CBAR    | <b>0.5140</b> (0.5001) | <b>0.9440</b> (0.0631) | <b>0.0580</b> (0.3909) | 0.4850(0.5441)         | <b>0.5430</b> (0.6592) | 0.0174(0.0561)         |
|     |     | LASSO   | 0.4480(0.4975)         | 0.9230(0.0859)         | 0.9190(1.2393)         | <b>0.0750</b> (0.2635) | 0.9940(1.2711)         | <b>0.0145</b> (0.0359) |
|     |     | ALASSO  | 0.2070(0.4054)         | 0.8741(0.0944)         | 0.7560(1.2908)         | 0.6460(0.5975)         | 1.4020(1.3031)         | 0.0207(0.0462)         |
|     |     | SCAD    | 0(0)                   | 0.7057(0.0972)         | 2.1690(2.4897)         | 1.3970(0.5897)         | 3.5660(2.1818)         | 0.0397(0.0693)         |
|     |     | LL-SCAD | 0.0010(0.0316)         | 0.6919(0.1084)         | 2.3350(2.5818)         | 1.4380(0.6516)         | 3.7730(2.3132)         | 0.5850(15.3729)        |
|     |     | MCP     | 0.0050(0.0706)         | 0.7338(0.1103)         | 2.1250(2.7053)         | 1.2240(0.5658)         | 3.3490(2.4814)         | 0.0330(0.0708)         |
|     |     | CBAR    | <b>0.7790</b> (0.4151) | <b>0.9768</b> (0.0436) | <b>0.0070</b> (0.0834) | 0.2140(0.4103)         | <b>0.2210</b> (0.4151) | 0.0198(0.0439)         |
|     |     | LASSO   | 0.5900(0.4921)         | 0.9528(0.0633)         | 0.5790(0.8405)         | <b>0.0080</b> (0.0891) | 0.5870(0.8421)         | <b>0.0171</b> (0.0440) |
| 50  |     | ALASSO  | 0.5690(0.4955)         | 0.9527(0.0579)         | 0.2270(0.5420)         | 0.2700(0.4442)         | 0.4970(0.6375)         | 0.0216(0.0459)         |
|     |     | SCAD    | 0.0330(0.1787)         | 0.8650(0.0621)         | 0.7550(1.2035)         | 0.7380(0.4512)         | 1.4930(0.9782)         | 0.0339(0.0438)         |
|     |     | LL-SCAD | 0.0360(0.1864)         | 0.8535(0.0750)         | 0.7540(1.2341)         | 0.8380(0.5330)         | 1.5920(1.0801)         | 0.1812(1.7996)         |
|     |     | MCP     | 0.0530(0.2241)         | 0.8763(0.0561)         | 0.6990(1.0861)         | 0.6650(0.4743)         | 1.3640(0.8463)         | 0.0321(0.0423)         |
|     |     | CBAR    | <b>0.7780</b> (0.4158) | <b>0.9764</b> (0.0445) | <b>0.0050</b> (0.0706) | 0.2190(0.4186)         | <b>0.2240</b> (0.4219) | 0.0092(0.0404)         |
|     |     | LASSO   | 0.5860(0.4928)         | 0.9482(0.0716)         | 0.6630(1.0385)         | <b>0.0090</b> (0.0945) | 0.6720(1.0380)         | <b>0.0079</b> (0.0334) |
|     |     | ALASSO  | 0.4800(0.4998)         | 0.9385(0.0663)         | 0.3350(0.7023)         | 0.3250(0.4729)         | 0.6600(0.7701)         | 0.0111(0.0451)         |
|     |     | SCAD    | 0.0020(0.0447)         | 0.8211(0.0852)         | 1.1060(1.5605)         | 0.9270(0.3190)         | 2.0330(1.4035)         | 0.0175(0.0490)         |
|     |     | LL-SCAD | 0.0140(0.1175)         | NaN(NA)                | 1.1840(1.6885)         | 0.9820(0.4711)         | 2.1660(1.5915)         | 0.0828(0.7559)         |
|     |     | MCP     | 0.0060(0.0773)         | 0.8447(0.0767)         | 0.8690(1.5113)         | 0.8820(0.3438)         | 1.7510(1.3209)         | 0.0166(0.0477)         |
| 80  |     | CBAR    | <b>0.7710</b> (0.4204) | <b>0.9757</b> (0.0449) | <b>0.0100</b> (0.1091) | 0.2220(0.4182)         | <b>0.2320</b> (0.4294) | 0.0054(0.0190)         |
|     |     | LASSO   | 0.5730(0.4949)         | 0.9467(0.0712)         | 0.6700(1.0011)         | <b>0.0130</b> (0.1133) | 0.6830(1.0038)         | <b>0.0045</b> (0.0111) |
|     |     | ALASSO  | 0.4100(0.4921)         | 0.9277(0.0702)         | 0.4220(0.8358)         | 0.3690(0.4848)         | 0.7910(0.8661)         | 0.0068(0.0176)         |
|     |     | SCAD    | 0.0010(0.0316)         | 0.7859(0.1000)         | 1.5280(1.8646)         | 1.0020(0.2934)         | 2.5300(1.7630)         | 0.0115(0.0206)         |
|     |     | LL-SCAD | 0.0090(0.0945)         | 0.7728(0.1158)         | 1.6350(2.0828)         | 1.0670(0.4783)         | 2.7020(2.0187)         | 0.0412(0.1478)         |
|     |     | MCP     | 0.0020(0.0447)         | 0.8247(0.0917)         | 1.0780(1.7467)         | 0.9440(0.2810)         | 2.0220(1.6156)         | 0.0101(0.0195)         |

**Table 6** Variable selection and estimation results (with standard deviation in parentheses) for all methods under Case II with  $\varepsilon \sim N(0, 1)$ ,  $U \sim N(5, 1)$  and fixed  $n = 400$ 

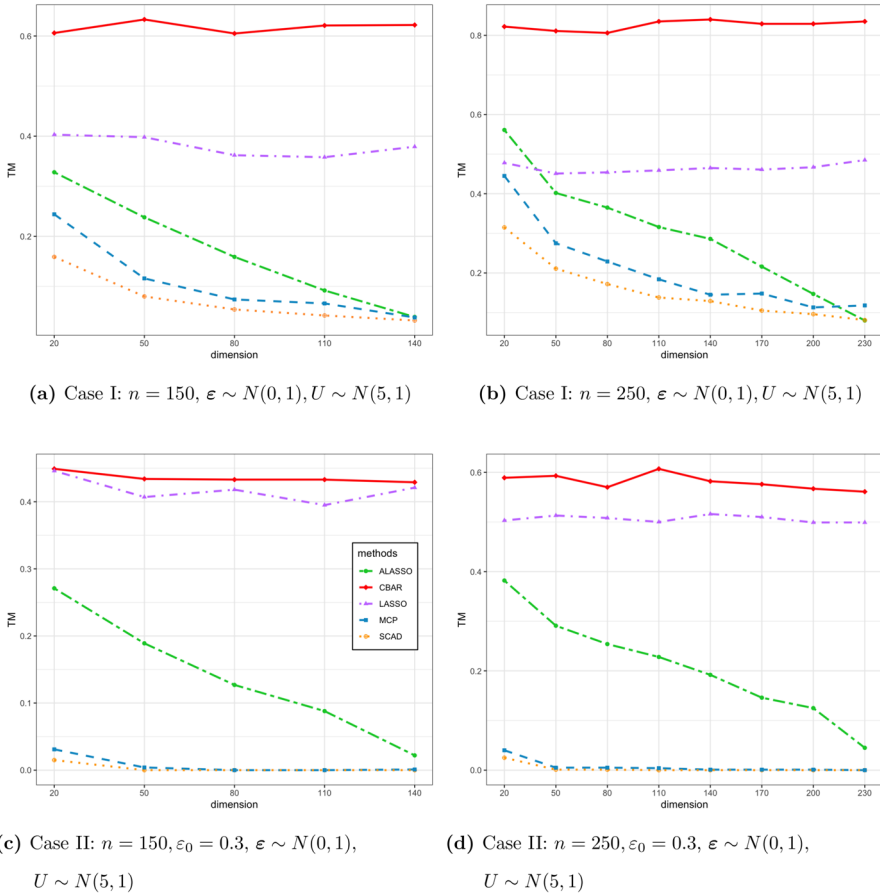
| $n$ | $p$ | Method  | TM                     | SM                     | FP                     | FN                     | Misc                   | MSE                    |
|-----|-----|---------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 400 | 100 | CBAR    | <b>0.7950</b> (0.4039) | <b>0.9783</b> (0.0429) | <b>0.0050</b> (0.0706) | 0.2010(0.4034)         | <b>0.2060</b> (0.4071) | 0.0043(0.0141)         |
|     |     | LASSO   | 0.5780(0.4941)         | 0.9457(0.0735)         | 0.6910(1.0414)         | <b>0.0110</b> (0.1043) | 0.7020(1.0470)         | <b>0.0037</b> (0.0102) |
|     |     | ALASSO  | 0.4070(0.4915)         | 0.9270(0.0713)         | 0.4270(0.8540)         | 0.3730(0.4900)         | 0.8000(0.8870)         | 0.0056(0.0132)         |
|     |     | SCAD    | 0(0)                   | 0.7751(0.1011)         | 1.6610(1.9067)         | 1.0210(0.3108)         | 2.6820(1.8039)         | 0.0098(0.0161)         |
|     |     | LL-SCAD | 0.0030(0.0547)         | 0.7585(0.1098)         | 1.8160(2.0670)         | 1.0920(0.4602)         | 2.9080(1.9701)         | 0.0457(0.3945)         |
| 200 | 200 | MCP     | 0.0010(0.0316)         | 0.8157(0.0986)         | 1.2450(1.9811)         | 0.9490(0.3073)         | 2.1940(1.8307)         | 0.0085(0.0149)         |
|     |     | CBAR    | <b>0.7760</b> (0.4171) | <b>0.9761</b> (0.0446) | <b>0.0030</b> (0.0547) | 0.2230(0.4188)         | <b>0.2260</b> (0.4232) | 0.0020(0.0048)         |
|     |     | LASSO   | 0.5550(0.4972)         | 0.9465(0.0691)         | 0.6710(0.9827)         | <b>0.0100</b> (0.0995) | 0.6810(0.9819)         | <b>0.0019</b> (0.0046) |
|     |     | ALASSO  | 0.2880(0.4530)         | 0.9051(0.0782)         | 0.5510(1.0717)         | 0.5000(0.5101)         | 1.0510(1.0703)         | 0.0032(0.0050)         |
|     |     | SCAD    | 0(0)                   | 0.7300(0.1115)         | 2.3610(2.4777)         | 1.1130(0.3826)         | 3.4740(2.3321)         | 0.0063(0.0077)         |
| 400 | 200 | LL-SCAD | 0(0)                   | 0.7213(0.1191)         | 2.4730(2.6624)         | 1.1490(0.4950)         | 3.6220(2.5182)         | 0.0163(0.0517)         |
|     |     | MCP     | 0.0030(0.0547)         | 0.7836(0.1134)         | 1.6890(2.4245)         | 1.0070(0.2847)         | 2.6960(2.3292)         | 0.0045(0.0058)         |

**Table 7** [Case II :  $p_n = 20, \varepsilon_0 = 0.3, \varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$ ] Parameter estimation and standard deviation (in parentheses) for all methods

| $(p_n, n)$ | Method  | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | $\hat{\beta}_4$ | $\hat{\beta}_5$ | $\hat{\sigma}^2$  |
|------------|---------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|
| (20,100)   | CBAR    | 3.9020(0.8243)  | 2.8692(0.7634)  | 0.6300(0.7358)  | 1.4565(0.8542)  | 0.9449(0.5191)  | 1.6134(1.1111)    |
|            | LASSO   | 3.6768(0.7384)  | 2.6045(0.6056)  | 0.7482(0.5792)  | 1.5543(0.6928)  | 0.7686(0.4052)  | 1.6577(1.1705)    |
|            | ALASSO  | 4.0632(0.8768)  | 2.9407(0.7565)  | 0.5489(0.7212)  | 1.3012(0.9004)  | 0.7693(0.5048)  | 1.6182(1.1376)    |
|            | SCAD    | 4.1673(1.0281)  | 3.0889(0.8520)  | 0.4232(0.8124)  | 1.1646(1.1294)  | 0.8830(0.5807)  | 1.6482(1.1649)    |
|            | LL-SCAD | 4.0295(1.7421)  | 3.0630(1.1866)  | 0.4095(1.0583)  | 1.3760(1.8480)  | 0.7711(0.9321)  | 28.0474(496.9494) |
|            | MCP     | 4.0786(0.9716)  | 3.0873(0.8067)  | 0.4225(0.7705)  | 1.2459(1.0651)  | 0.9576(0.5627)  | 1.6149(1.1459)    |
| (20,200)   | NV-OLS  | 3.8004(0.7390)  | 2.6783(0.6440)  | 0.8234(0.6194)  | 1.5555(0.7154)  | 1.0418(0.4073)  | 1.6177(1.1785)    |
|            | CBAR    | 3.9136(0.5561)  | 3.0376(0.5141)  | 0.4716(0.4945)  | 1.5223(0.5279)  | 0.9944(0.2847)  | 1.3330(0.7172)    |
|            | LASSO   | 3.8000(0.5328)  | 2.8330(0.4013)  | 0.5742(0.3786)  | 1.5512(0.4877)  | 0.8360(0.2665)  | 1.3571(0.7443)    |
|            | ALASSO  | 4.0097(0.5867)  | 3.1309(0.4781)  | 0.3764(0.4552)  | 1.4314(0.5677)  | 0.8963(0.3076)  | 1.3355(0.7326)    |
|            | SCAD    | 3.9711(0.6460)  | 3.2834(0.5275)  | 0.2299(0.5023)  | 1.4603(0.6805)  | 0.9813(0.3298)  | 1.3485(0.7451)    |
|            | LL-SCAD | 3.8827(1.1853)  | 3.2187(0.8428)  | 0.2837(0.7563)  | 1.5978(1.2712)  | 0.9161(0.4432)  | 2.9519(24.1422)   |
| (20,400)   | MCP     | 3.9418(0.6133)  | 3.2608(0.5298)  | 0.2515(0.5062)  | 1.4862(0.6268)  | 1.0075(0.3099)  | 1.3392(0.7364)    |
|            | NV-OLS  | 3.8886(0.5316)  | 2.8644(0.4082)  | 0.6448(0.3862)  | 1.5474(0.4982)  | 1.0205(0.2766)  | 1.3309(0.7245)    |
|            | CBAR    | 3.9529(0.2865)  | 2.9907(0.3243)  | 0.5173(0.3153)  | 1.5429(0.2915)  | 1.0028(0.1389)  | 1.1542(0.2167)    |
|            | LASSO   | 3.8807(0.2855)  | 2.8904(0.2347)  | 0.5463(0.2239)  | 1.5550(0.2863)  | 0.8854(0.1405)  | 1.1661(0.2198)    |
|            | ALASSO  | 4.0066(0.3018)  | 3.1146(0.3068)  | 0.3937(0.2991)  | 1.4909(0.3137)  | 0.9529(0.1486)  | 1.1556(0.2182)    |
|            | SCAD    | 3.9458(0.2935)  | 3.2830(0.3907)  | 0.2267(0.3852)  | 1.5510(0.3044)  | 1.0070(0.1506)  | 1.1668(0.2190)    |
| (20,600)   | LL-SCAD | 3.8750(1.0804)  | 3.2449(0.6572)  | 0.2671(0.6205)  | 1.6162(1.1133)  | 0.9944(0.3190)  | 1.7258(3.6901)    |
|            | MCP     | 3.9438(0.2843)  | 3.2346(0.4024)  | 0.2751(0.3981)  | 1.5516(0.2889)  | 1.0127(0.1418)  | 1.1618(0.2194)    |
|            | NV-OLS  | 3.9422(0.2830)  | 2.9057(0.2348)  | 0.6022(0.2232)  | 1.5533(0.2857)  | 1.0129(0.1374)  | 1.1528(0.2160)    |
|            | CBAR    | 3.9984(0.1729)  | 2.9688(0.2258)  | 0.5442(0.2159)  | 1.5017(0.1868)  | 1.0157(0.1041)  | 1.0920(0.1076)    |
|            | LASSO   | 3.9426(0.1730)  | 2.9179(0.1749)  | 0.5381(0.1645)  | 1.5094(0.1855)  | 0.9190(0.1038)  | 1.1071(0.1085)    |
|            | ALASSO  | 4.0387(0.1801)  | 3.0970(0.2336)  | 0.4172(0.2236)  | 1.4619(0.1979)  | 0.9792(0.1092)  | 1.1005(0.1080)    |
| (20,800)   | SCAD    | 3.9930(0.1729)  | 3.2180(0.3697)  | 0.2967(0.3619)  | 1.5073(0.1863)  | 1.0201(0.1041)  | 1.1087(0.1098)    |
|            | LL-SCAD | 3.9121(0.7229)  | 3.1832(0.5900)  | 0.3326(0.5508)  | 1.5915(0.7401)  | 0.9924(0.2381)  | 1.8175(8.4755)    |
|            | MCP     | 3.9928(0.1723)  | 3.1599(0.3647)  | 0.3547(0.3583)  | 1.5074(0.1858)  | 1.0202(0.1048)  | 1.1042(0.1107)    |
|            | NV-OLS  | 3.9924(0.1718)  | 2.9286(0.1745)  | 0.5844(0.1632)  | 1.5075(0.1849)  | 1.0214(0.1025)  | 1.0985(0.1074)    |

**Table 8** [Case II :  $n = 400, \varepsilon_0 = 0.3, \varepsilon \sim N(0, 1)$  and  $U \sim N(5, 1)$ ] Parameter estimation and standard deviation (in parentheses) for all methods

| $(p_n, n)$ | Method  | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | $\hat{\beta}_4$ | $\hat{\beta}_5$ | $\hat{\sigma}^2$ |
|------------|---------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
| (50, 400)  | CBAR    | 3.9542(0.3550)  | 3.0083(0.3146)  | 0.4995(0.3041)  | 1.5340(0.3326)  | 0.9992(0.1624)  | 1.1733(0.3788)   |
|            | LASSO   | 3.8639(0.3437)  | 2.8927(0.2222)  | 0.5224(0.2103)  | 1.5452(0.3091)  | 0.8505(0.1578)  | 1.1951(0.3835)   |
|            | ALASSO  | 4.0283(0.3842)  | 3.1778(0.2905)  | 0.3301(0.2805)  | 1.4627(0.3699)  | 0.9287(0.1705)  | 1.1758(0.3751)   |
|            | SCAD    | 3.9544(0.3948)  | 3.4263(0.2852)  | 0.0845(0.2722)  | 1.5391(0.3988)  | 0.9889(0.1875)  | 1.1945(0.3709)   |
|            | LL-SCAD | 3.9042(0.8559)  | 3.3194(0.6894)  | 0.1801(0.6318)  | 1.5995(0.9038)  | 0.9441(0.3604)  | 4.6078(72.2517)  |
|            | MCP     | 3.9462(0.3778)  | 3.3975(0.3148)  | 0.1129(0.3029)  | 1.5427(0.3654)  | 1.0062(0.1651)  | 1.1885(0.3680)   |
| (80, 400)  | NV-OLS  | 3.9408(0.3674)  | 2.9145(0.2231)  | 0.5930(0.2106)  | 1.5477(0.3452)  | 1.0103(0.1574)  | 1.1711(0.3711)   |
|            | CBAR    | 3.9658(0.3204)  | 3.0189(0.3177)  | 0.4923(0.3059)  | 1.5200(0.3280)  | 1.0007(0.1578)  | 1.1686(0.3494)   |
|            | LASSO   | 3.8636(0.3024)  | 2.8963(0.2266)  | 0.5122(0.2132)  | 1.5347(0.2978)  | 0.8367(0.1502)  | 1.1977(0.3706)   |
|            | ALASSO  | 4.0541(0.3394)  | 3.2168(0.2944)  | 0.2942(0.2792)  | 1.4343(0.3486)  | 0.9190(0.1679)  | 1.1736(0.3553)   |
|            | SCAD    | 3.9902(0.3934)  | 3.4718(0.2248)  | 0.0429(0.2081)  | 1.4970(0.4404)  | 0.9880(0.1966)  | 1.1958(0.3586)   |
|            | LL-SCAD | 3.9652(0.8592)  | 3.3669(0.7109)  | 0.1454(0.6706)  | 1.5237(0.9111)  | 0.9483(0.3187)  | 3.8125(62.5300)  |
| (100, 400) | MCP     | 3.9611(0.3365)  | 3.4469(0.2657)  | 0.0668(0.2489)  | 1.5248(0.3505)  | 1.0084(0.1656)  | 1.1869(0.3578)   |
|            | NV-OLS  | 3.9527(0.3175)  | 2.9212(0.2260)  | 0.5897(0.2138)  | 1.5328(0.3222)  | 1.0144(0.1561)  | 1.1676(0.3517)   |
|            | CBAR    | 3.9815(0.2987)  | 2.9934(0.3271)  | 0.5189(0.3147)  | 1.5064(0.3404)  | 1.0031(0.1483)  | 1.1584(0.2672)   |
|            | LASSO   | 3.8717(0.2845)  | 2.8738(0.2423)  | 0.5317(0.2273)  | 1.5267(0.2983)  | 0.8317(0.1457)  | 1.1904(0.3034)   |
|            | ALASSO  | 4.0789(0.3194)  | 3.2063(0.3130)  | 0.3058(0.2992)  | 1.4115(0.3700)  | 0.9142(0.1673)  | 1.1647(0.2703)   |
|            | SCAD    | 4.0187(0.3935)  | 3.4728(0.2367)  | 0.0438(0.2203)  | 1.4705(0.4882)  | 0.9823(0.2065)  | 1.1898(0.2761)   |
| (100, 400) | LL-SCAD | 3.9370(1.2288)  | 3.3651(0.6216)  | 0.1424(0.5518)  | 1.5775(1.3445)  | 0.9191(0.2992)  | 2.6534(23.0005)  |
|            | MCP     | 3.9861(0.3329)  | 3.4376(0.2920)  | 0.0779(0.2777)  | 1.5009(0.4021)  | 1.0079(0.1676)  | 1.1782(0.2738)   |
|            | NV-OLS  | 3.9676(0.2949)  | 2.9018(0.2444)  | 0.6103(0.2297)  | 1.5200(0.3359)  | 1.0153(0.1460)  | 1.1570(0.2674)   |



**Fig. 1** The TM results of all methods across various  $p_n$  for Case I and Case II

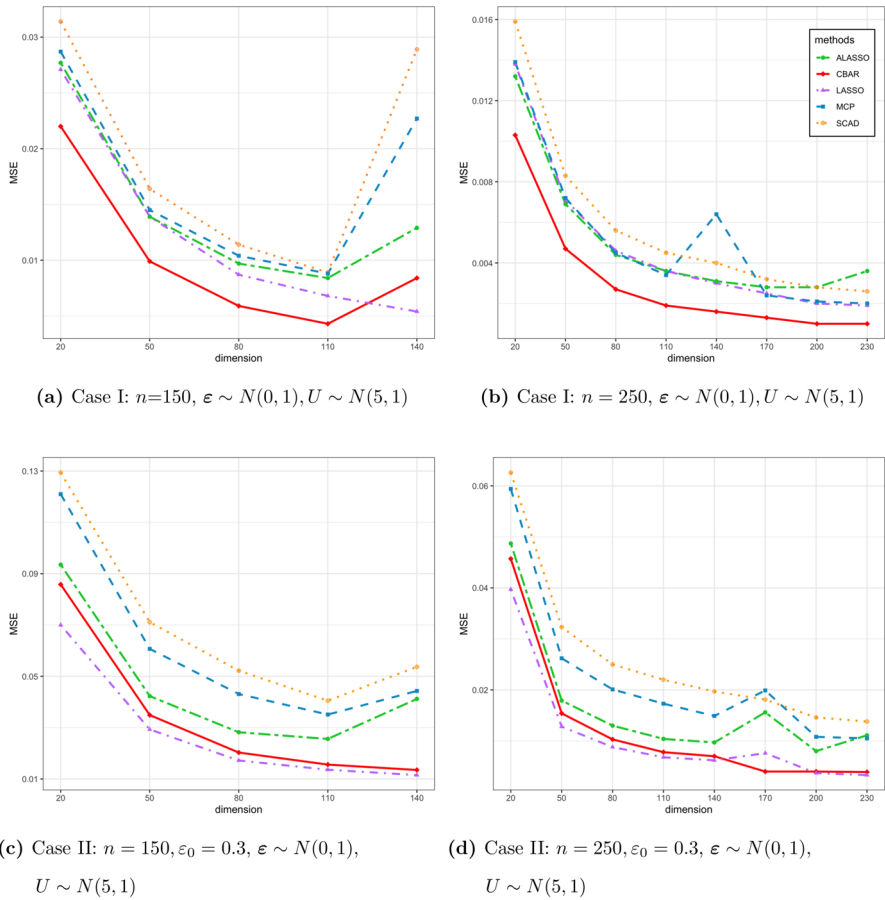
We have discussed the estimation of regression coefficients above. Next, we want to conduct hypothesis test on if certain explanatory variables influence the response significantly. We consider the following hypothesis test problem:

$$\mathcal{H}_{j0} : \beta_j = 0 \quad \text{v.s.} \quad \mathcal{H}_{j1} : \beta_j \neq 0, \quad j \in \{j : \hat{\beta}_j^* \neq 0, j = 1, 2, \dots, 12\}.$$

Applying Theorem 1, we can construct our test statistic as

$$\mathcal{T}_{nj} = \left| \sqrt{n} \hat{s}_n^{-1} \mathbf{b}_n^T (\hat{\beta}_1^* - \beta_{01}) \right|, \quad (15)$$

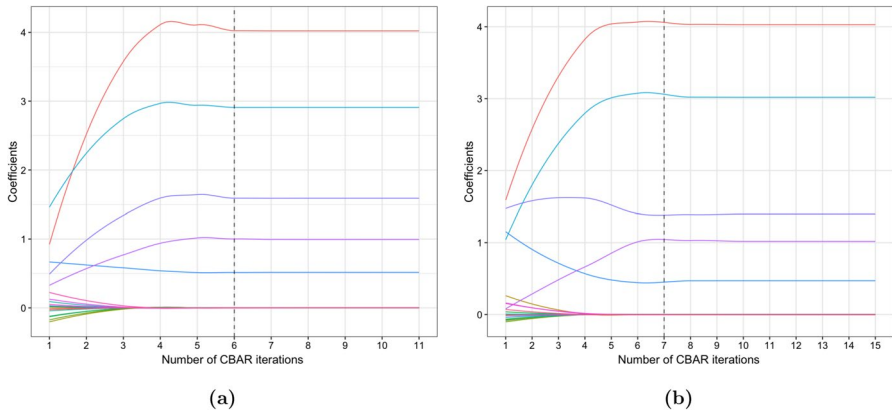
where  $\hat{s}_n = \sqrt{\mathbf{b}_n^T (\hat{\sigma}^2 \hat{\Omega}_1 + \hat{\Gamma}_n) \mathbf{b}_n}$ ,  $\mathbf{b}_n$  is a 5-dimensional vector whose  $j$ th element is 1 and others are 0.  $\hat{\beta}_1^*$  is a 5-dimensional vector composed of the non-zero elements in the CBAR estimator. If the value of  $\mathcal{T}_n$  is large, it indicates that the null hypothesis



**Fig. 2** The MSE results of all methods across various  $p_n$  for Case I and Case II

**Table 9** (Grouping effects) Estimated coefficients of CBAR, Elastic Net, LASSO, Adaptive LASSO, SCAD and MCP with optimal tuning parameters selected by BIC

| Parameter | CBAR               | Elastic Net         | LASSO              | ALASSO             | SCAD               | MCP                |
|-----------|--------------------|---------------------|--------------------|--------------------|--------------------|--------------------|
| $\beta_1$ | 1.9603             | 1.9872              | 2.9638             | 1.1823             | 2.3815             | 3.0831             |
| $\beta_2$ | 2.0472             | 1.9852              | 4.0905             | 1.9141             | 2.5705             | 0                  |
| $\beta_3$ | 2.0356             | 1.9895              | -1.1664            | 2.7761             | 0.7948             | 3.3117             |
| $\beta_4$ | 0.5144             | 0.5296              | 0                  | 0                  | -0.5869            | 1.4103             |
| $\beta_5$ | 0.4687             | 0.4918              | 0                  | 1.6478             | 2.1933             | 0                  |
| $\beta_6$ | 0.5356             | 0.4408              | 1.4622             | 0                  | 0                  | 0                  |
| $\beta_7$ | 0                  | 0                   | -0.0032            | 0                  | 0                  | 0.0667             |
| $\beta_8$ | 0                  | 0                   | 0.0026             | 0                  | 0                  | -0.1060            |
| $\beta_9$ | 0                  | 0                   | -0.0295            | 0                  | 0                  | 0                  |
| Tuning    | $\xi_n = 100$      | $\lambda_1 = 0.022$ | $\lambda = 0.0043$ | $\gamma = 1$       | $\gamma = 3.7$     | $\gamma = 3$       |
| Parameter | $\lambda_n = 0.05$ | $\lambda_2 = 0.001$ |                    | $\lambda = 0.7005$ | $\lambda = 0.0181$ | $\lambda = 0.0107$ |



**Fig. 3** The iterative process of the CBAR algorithm for two different simulation scenarios. (a). Case I:  $n = 100, p_n = 20, \varepsilon \sim N(0, 1), U \sim N(5, 1)$ ; (b). Case II:  $n = 100, p_n = 20, \varepsilon_0 = 0.3, \varepsilon \sim N(0, 1), U \sim N(5, 1)$

**Table 10** Correlation coefficients of PTRATIO with (a) the observable response and other predictors (without accounting for its distorting effect). (b) the recovered response and predictors (with PTRATIO taken as a confounding covariate)

| Variable   | Without considering distortion | Considering distortion  |
|------------|--------------------------------|-------------------------|
| MV         | -0.5078                        | -0.0036                 |
| RM         | -0.3555                        | -0.0104                 |
| AGE        | 0.2615                         | 0.0180                  |
| log(DIS)   | -0.2325                        | $-5.78 \times 10^{-16}$ |
| log(RAD)   | 0.4647                         | $1.39 \times 10^{-16}$  |
| TAX        | 0.4608                         | 0.0236                  |
| B          | -0.1774                        | -0.0039                 |
| log(LSTAT) | 0.3740                         | 0.0189                  |
| CRIM       | 0.2899                         | 0.0203                  |
| ZN         | -0.3917                        | $-4.72 \times 10^{-16}$ |
| INDUS      | 0.3832                         | $-1.94 \times 10^{-16}$ |
| CHAS       | -0.1215                        | 0.0016                  |
| NOX        | 0.1889                         | $-5.37 \times 10^{-16}$ |

$\mathcal{H}_0$  should be rejected. Therefore, we reject the null hypothesis if  $\mathcal{T}_{nj} \geq Z_{1-\alpha/2}$ , where  $Z_{1-\alpha/2}$  is the  $100(1 - \alpha/2)\%$  quantile of standard normal distribution,  $\alpha$  is the significance level.

From Table 11, the fitted model, with  $\mathcal{T}_{nj}$  values given in parentheses, is as follows:

$$\begin{aligned}
 \widehat{\log(\text{MV})} = & 4.0401 + 0.0035\text{RM}^2 - 0.3804\log(\text{LSTAT}) - 0.0087\text{CRIM} \\
 & (5.9420) \quad (22.6076) \quad (12.2308) \\
 & + 0.2539\text{CHAS} - 0.5701\text{NOX}^2 \\
 & (9.3001) \quad (21.3508)
 \end{aligned} \quad (16)$$

The rejection region is  $\{\mathcal{T}_{nj} > 2.575\}$  at the 0.01 level of significance. The  $\mathcal{T}_{nj}$ -values (in parentheses) in Equation (16) indicate that, at significance level 0.01, the

**Table 11** Estimated coefficients of CBAR, LASSO, ALASSO, SCAD and MCP and BIC values for the Boston Housing data considering the covariate  $U$ 

| Variables(parameter)                        | CBAR                 | LASSO                | ALASSO             | SCAD                | MCP               |
|---------------------------------------------|----------------------|----------------------|--------------------|---------------------|-------------------|
| Intercept( $\hat{\beta}_0$ )                | 4.0401               | 4.0000               | 4.0881             | 4.1794              | 4.2280            |
| RM <sup>2</sup> ( $\hat{\beta}_1$ )         | 0.0035               | 0.0031               | 0.0023             | 0.0040              | 0                 |
| AGE( $\hat{\beta}_2$ )                      | 0                    | 0                    | 0                  | 0.0006              | 0                 |
| log(DIS)( $\hat{\beta}_3$ )                 | 0                    | 0                    | 0                  | -0.0542             | 0                 |
| log(RAD)( $\hat{\beta}_4$ )                 | 0                    | 0.0214               | 0.0532             | 0.1013              | 0                 |
| TAX( $\hat{\beta}_5$ )                      | 0                    | -0.0002              | -0.0003            | -0.0005             | 0                 |
| (B - 0.63) <sup>2</sup> ( $\hat{\beta}_7$ ) | 0                    | 0                    | 0                  | 0                   | 0                 |
| log(LSTAT)( $\hat{\beta}_8$ )               | -0.3804              | -0.3724              | -0.3947            | -0.3839             | -0.4365           |
| CRIM( $\hat{\beta}_9$ )                     | -0.0087              | -0.0068              | -0.0078            | -0.0099             | -0.0070           |
| ZN( $\hat{\beta}_{10}$ )                    | 0                    | 0                    | 0                  | -0.0016             | 0                 |
| INDUS( $\hat{\beta}_{11}$ )                 | 0                    | 0                    | 0                  | 0                   | 0                 |
| CHAS( $\hat{\beta}_{12}$ )                  | 0.2539               | 0.1907               | 0.2170             | 0.2354              | 0.2056            |
| NOX <sup>2</sup> ( $\hat{\beta}_{13}$ )     | -0.5701              | -0.4155              | -0.4813            | -0.9217             | -0.2987           |
| Tuning                                      | $\xi_n = 80.8081$    | $\lambda_n = 0.0005$ | $\gamma=1$         | $\gamma=3.7$        | $\gamma=3$        |
| Parameter                                   | $\lambda_n = 0.0203$ |                      | $\lambda_n=0.0006$ | $\lambda_n= 0.0025$ | $\lambda= 0.0186$ |
| Number of selected                          | 5                    | 7                    | 7                  | 10                  | 4                 |
| BIC                                         | -3.1593              | -3.1316              | -3.1433            | -3.1381             | -3.1498           |

significant variables selected by CBAR method exactly have significant effect on the MV, and this aligns with closely the findings of (Zhang et al., 2012a). The fitted model (16) reflects the refinement of the method. In our analysis, our initial model, which employed an inappropriate bandwidth range, selected 8 predictors with a BIC of -3.2055. In contrast, the refined model, using a more appropriate bandwidth range, yields a more parsimonious model with only 5 predictors. Moreover, using fewer active features, CBAR achieves a comparable BIC value of -3.1593. The difference of 0.0462 is negligible. Therefore, CBAR method is more attractive than its competitors for exploring true relationship between these factors and house price.

## 5 Conclusion

In this paper, we have extended the BAR penalized method for simultaneous estimation and variable selection to the linear CAR model, where the response and predictors are distorted in a multiplicative fashion and cannot be directly observed. To remove the distortion, we use non-parametric regression to obtain consistent estimators of the unknown distortion functions  $\psi(U)$  and  $\phi_j(U)$  and accordingly obtain the estimates  $\hat{Y}$  and  $\hat{X}$  of the unobservable real response and predictors. Then naturally, we can estimate the coefficients  $\beta$  based on  $\hat{Y}$  and  $\hat{X}$  using our CBAR method.

We have established that the CBAR procedure is asymptotically consistent for variable selection, has an oracle estimation property and possesses a grouping property under high-dimensional setting. Furthermore, we have demonstrated that  $\hat{\sigma}^2$  serves as a

consistent estimator for  $\sigma^2$ . The simulations indicate that our CBAR procedure is much superior to the LASSO, Adaptive LASSO, SCAD, and MCP methods in both variable selection and coefficient estimation, particularly when variables are highly correlated. Furthermore, the CBAR algorithm enjoys a rapid convergence rate. In application to Boston Housing data, the selected significant variables by CBAR are exactly having significant effect on response. Hence, we strongly recommend the CBAR procedure to select variables and estimate parameters simultaneously for other various models.

## Appendix

We first introduce some notations. Denote  $\mathbf{S}_n = n^{-1}\mathbf{X}^T\mathbf{X}$  and  $\mathbf{V}_n = n^{-1}\hat{\mathbf{X}}^T\hat{\mathbf{X}}$  and write

$$\begin{pmatrix} \alpha(\beta) \\ \gamma(\beta) \end{pmatrix} \equiv g(\beta) = \left( \hat{\mathbf{X}}^T\hat{\mathbf{X}} + \lambda_n \mathbf{D}^{-1}(\beta) \right)^{-1} \hat{\mathbf{X}}^T\hat{\mathbf{Y}}. \quad (\text{A.1})$$

We partition  $\mathbf{V}_n^{-1}$  and  $\mathbf{D}^{-1}(\beta)$  into

$$\mathbf{V}_n^{-1} = \left( n^{-1}\hat{\mathbf{X}}^T\hat{\mathbf{X}} \right)^{-1} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{12}^T & \mathbf{A}_{22} \end{pmatrix} \quad \text{and} \quad \mathbf{D}^{-1}(\beta) = \begin{pmatrix} \mathbf{D}_1^{-1}(\beta_1) & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_2^{-1}(\beta_2) \end{pmatrix},$$

where  $\mathbf{A}_{11}$  is a  $q_n \times q_n$  matrix,  $\mathbf{D}_1(\beta_1) = \text{diag}(\beta_1^2, \dots, \beta_{q_n}^2)$  and  $\mathbf{D}_2(\beta_2) = \text{diag}(\beta_{q_n+1}^2, \dots, \beta_{p_n}^2)$ . The matrix  $\mathbf{V}_n^{-1}$  exists with probability 1 as  $n$  is sufficiently large. Then multiplying both sides of (A.1) by  $(\hat{\mathbf{X}}^T\hat{\mathbf{X}})^{-1}(\hat{\mathbf{X}}^T\hat{\mathbf{X}} + \lambda_n \mathbf{D}^{-1}(\beta))$ , we have

$$\begin{pmatrix} \alpha(\beta) - \beta_{01} \\ \gamma(\beta) \end{pmatrix} + \frac{\lambda_n}{n} \begin{pmatrix} \mathbf{A}_{11}\mathbf{D}_1^{-1}(\beta_1)\alpha(\beta) + \mathbf{A}_{12}\mathbf{D}_2^{-1}(\beta_2)\gamma(\beta) \\ \mathbf{A}_{12}^T\mathbf{D}_1^{-1}(\beta_1)\alpha(\beta) + \mathbf{A}_{22}\mathbf{D}_2^{-1}(\beta_2)\gamma(\beta) \end{pmatrix} = (\hat{\mathbf{X}}^T\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}^T\hat{\mathbf{Y}} - \beta_{0n}. \quad (\text{A.2})$$

Then, we present some preliminary results needed to prove Theorem 1-3 in the Supplementary material for saving space.

**Proof of Proposition 1** We represent  $\hat{\beta}_{ols}$  as  $(\mathbf{S}_n + \Delta_{\mathbf{X}})^{-1}(n^{-1}\mathbf{X}^T\mathbf{Y} + \Delta_{(\mathbf{X}, \mathbf{Y})})$ , where  $\Delta_{(\mathbf{X}, \mathbf{Y})} = n^{-1}\hat{\mathbf{X}}^T\hat{\mathbf{Y}} - n^{-1}\mathbf{X}^T\mathbf{Y}$  and  $\Delta_{\mathbf{X}} = n^{-1}\hat{\mathbf{X}}^T\hat{\mathbf{X}} - n^{-1}\mathbf{X}^T\mathbf{X}$ . Then

$$\begin{aligned} & \hat{\beta}_{ols} - \beta_{0n} \\ &= (\mathbf{S}_n^{-1} - \mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1})(\mathbf{S}_n\beta_{0n} + n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon} + \Delta_{(\mathbf{X}, \mathbf{Y})}) - \beta_{0n} \\ &= n^{-1}\mathbf{S}_n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon} + \mathbf{S}_n^{-1}\Delta_{(\mathbf{X}, \mathbf{Y})} - \mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\mathbf{S}_n\beta_{0n} \\ & \quad - n^{-1}\mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon} - \mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\Delta_{(\mathbf{X}, \mathbf{Y})} \end{aligned} \quad (\text{A.3})$$

By condition (C6),  $E\|\frac{1}{n}\mathbf{X}^T\boldsymbol{\varepsilon}\|^2 = O_p(p_n n^{-1})$ , which yields that  $\|n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\| = O_p(p_n^{1/2} n^{-1/2})$ . Combining Lemma 5 with condition (C6) gives  $\|\mathbf{S}_n\| = O_p(1)$ ,  $\|\mathbf{V}_n\| = O_p(1)$  and  $\|\mathbf{S}_n^{-1}\| = O_p(1)$ . Hence

$$\|n^{-1}\mathbf{S}_n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\| \leq \|\mathbf{S}_n^{-1}\| \|n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\| = O_p(p_n^{1/2} n^{-1/2}). \quad (\text{A.4})$$

By Corollary 1, it is straightforward to show that  $\|\mathbf{S}_n^{-1}\Delta_{(\mathbf{X}, \mathbf{Y})}\| \leq \|\mathbf{S}_n^{-1}\| \|\Delta_{(\mathbf{X}, \mathbf{Y})}\| = O_p(a_{1n} p_n n^{-1/2})$ . By  $a_{1n} p_n n^{-1/2} \rightarrow 0$  implied by condition (C7), we have

$$\|\mathbf{S}_n^{-1}\Delta_{(\mathbf{X}, \mathbf{Y})}\| \rightarrow 0. \quad (\text{A.5})$$

Similar to the analysis above, by condition (C7), we have  $\|n^{-1}\mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\| = O_p(p_n^{1/2} n^{-1/2}) \rightarrow 0$ ,  $\|\mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\Delta_{(\mathbf{X}, \mathbf{Y})}\| = O_p(a_{1n} p_n^{3/2} n^{-1}) \rightarrow 0$  and  $\|\mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\mathbf{S}_n\boldsymbol{\beta}_{0n}\| = O_p(p_n n^{-1/2})$ .

This, together with (A.3)-(A.5) and  $p_n^2/n \rightarrow 0$  yields

$$\begin{aligned} \|\hat{\boldsymbol{\beta}}_{ols} - \boldsymbol{\beta}_{0n}\| &= \|-\mathbf{S}_n^{-1}\Delta_{\mathbf{X}}\mathbf{V}_n^{-1}\mathbf{S}_n\boldsymbol{\beta}_{0n}\| + o_p(1) \\ &= O_p(p_n n^{-1/2}). \end{aligned} \quad (\text{A.6})$$

We complete the proof of Proposition 1. □

**Proof of Proposition 2** It can be shown that

$$\hat{\boldsymbol{\beta}}^{(0)} - \boldsymbol{\beta}_{0n} = \left\{ \left( \mathbf{S}_n + \Delta_{\mathbf{X}} + \frac{\xi_n}{n} \mathbf{I}_{p_n} \right)^{-1} (\Delta_{(\mathbf{X}, \mathbf{Y})} + n^{-1} \mathbf{X}^T \mathbf{Y}) \right\} - \boldsymbol{\beta}_{0n},$$

where  $\Delta_{\mathbf{X}}$  and  $\Delta_{(\mathbf{X}, \mathbf{Y})}$  defined as in the proof of Proposition 1. Similar to the proof of Proposition 1, we have that

$$\left( \mathbf{S}_n + \Delta_{\mathbf{X}} + \frac{\xi_n}{n} \mathbf{I}_{p_n} \right)^{-1} = \mathbf{S}_n^{-1} - \mathbf{S}_n^{-1} \left( \Delta_{\mathbf{X}} + \frac{\xi_n}{n} \mathbf{I}_{p_n} \right) \mathbf{S}_n^{-1} + O_p(p_n^{1/2} n^{-1/2}),$$

according to this and by some simple calculations, we have

$$\hat{\boldsymbol{\beta}}^{(0)} - \boldsymbol{\beta}_{0n} = n^{-1} \mathbf{S}_n^{-1} \mathbf{X}^T \boldsymbol{\varepsilon} - \mathbf{S}_n^{-1} \left( \Delta_{\mathbf{X}} + \frac{\xi_n}{n} \mathbf{I}_{p_n} \right) \boldsymbol{\beta}_{0n} + o_p(1),$$

thus  $\|\hat{\boldsymbol{\beta}}^{(0)} - \boldsymbol{\beta}_{0n}\| = O_p(\sqrt{p_n^2/n})$ . This completes the proof. □

**Lemma 1** Let  $\delta_n$  be a sequence of positive real numbers satisfying  $\delta_n \rightarrow \infty$  and  $p_n^2 \delta_n^2 / \lambda_n \rightarrow 0$ . Define  $\mathbf{H}_n \equiv \{\boldsymbol{\beta} = (\boldsymbol{\beta}_1^T, \boldsymbol{\beta}_2^T)^T : \|\boldsymbol{\beta}_1 - \boldsymbol{\beta}_{01}\| \leq \delta_n \sqrt{p_n^2/n}, \|\boldsymbol{\beta}_2\| \leq \delta_n \sqrt{p_n^2/n}\}$ , where  $\boldsymbol{\beta}_1 \in \mathbb{R}^{q_n}, \boldsymbol{\beta}_2 \in \mathbb{R}^{p_n - q_n}$ . Then

under the conditions (a), (b) and (C1)-(C7) and with probability tending to 1, we have that

(a) For the second part of  $g(\beta)$

$$\sup_{\beta \in \mathbf{H}_n} \frac{\|\gamma(\beta)\|}{\|\beta_2\|} < \frac{1}{C_0}, \quad \text{for some constant } C_0 > 1;$$

where  $\gamma(\beta)$  is defined as in (A.2).

(b) The function  $g(\cdot)$  is a mapping from  $\mathbf{H}_n$  to itself.

The proof of this Lemma is similar as that of Lemma 1 in (Dai et al., 2018) and is therefore omitted here.

**Lemma 2** Define transformation

$$f(\beta_1) = \left\{ \hat{\mathbf{X}}_I^T \hat{\mathbf{X}}_I + \lambda_n \mathbf{D}_1^{-1}(\beta_1) \right\}^{-1} \hat{\mathbf{X}}_I^T \hat{\mathbf{Y}}, \quad (\text{A.7})$$

where  $\mathbf{D}_1(\beta_1) = \text{diag}(\beta_1^2, \beta_2^2, \dots, \beta_{q_n}^2)$ . Assume that conditions (a), (b) and (C1)-(C7) hold, then we have,

- (a) with probability tending to 1,  $f(\beta_1)$  is a contraction mapping from  $\mathbf{B}_n \equiv \{\beta_1 \in \mathbb{R}^{q_n} : \|\beta_1 - \beta_{01}\| \leq \delta_n \sqrt{q_n/n}\}$  to itself;
- (b)  $\sqrt{n} s_n^{-1} \mathbf{b}_n^T (\hat{\beta}_1^\circ - \beta_{01}) \xrightarrow{D} \mathcal{N}(0, 1)$ , for any  $q_n$ -vector  $\mathbf{b}_n$  satisfying  $0 < \|\mathbf{b}_n\| \leq 1$ , define  $s_n^2 = \mathbf{b}_n^T (\sigma^2 \mathbf{\Omega}_1^{-1} + \mathbf{\Gamma}_n) \mathbf{b}_n$  as in Theorem 1, where  $\hat{\beta}_1^\circ$  is the unique fixed point of  $f(\beta_1)$  defined by  $\hat{\beta}_1^\circ = f(\hat{\beta}_1^\circ)$ .

**Proof of Lemma 2** In CAR framework, the contraction mapping property holds in a more restricted set as compared to that in Dai et al. (2018), due to the estimation

error in the recovered data  $(\hat{\mathbf{Y}}, \hat{\mathbf{X}})$ . The proof of the Lemma 2(a) is similar as that of Lemma 2(a) in (Dai et al., 2018) and is therefore omitted here. We only prove part (b).

Let  $\Delta_{\mathbf{X}_I} = n^{-1} \hat{\mathbf{X}}_I^T \hat{\mathbf{X}}_I - n^{-1} \mathbf{X}_I^T \mathbf{X}_I = O_p(q_n^{1/2} n^{-1/2})$  and  $\Delta_{(\mathbf{X}_I, \mathbf{Y})} = n^{-1} \hat{\mathbf{X}}_I^T \hat{\mathbf{Y}} - n^{-1} \mathbf{X}_I^T \mathbf{Y} = O_p(a_{1n} q_n n^{-1/2})$ . Thus,

$$\hat{\beta}_1^\circ = \left( \mathbf{S}_{11n} + \Delta_{\mathbf{X}_I} + \frac{\lambda_n}{n} \mathbf{D}_1^{-1}(\hat{\beta}_1^\circ) \right)^{-1} (n^{-1} \mathbf{X}_I^T \mathbf{Y} + \Delta_{(\mathbf{X}_I, \mathbf{Y})}).$$

Using a similar analysis to  $\hat{\beta}_{ols}$ , we have

$$\hat{\beta}_1^\circ = \left\{ \mathbf{S}_{11n}^{-1} - \mathbf{S}_{11n}^{-1} \left( \Delta_{\mathbf{X}_I} + \frac{\lambda_n}{n} \mathbf{D}_1^{-1}(\hat{\beta}_1^\circ) \right) \left( \mathbf{V}_{11n} + \frac{\lambda_n}{n} \mathbf{D}_1^{-1}(\hat{\beta}_1^\circ) \right)^{-1} \right\} (n^{-1} \mathbf{X}_I^T \mathbf{Y} + \Delta_{(\mathbf{X}_I, \mathbf{Y})}).$$

By some careful calculations, we have

$$\begin{aligned} n^{1/2}(\hat{\beta}_1^\circ - \beta_{01}) &= n^{-1/2} \mathbf{S}_{11n}^{-1} \mathbf{X}_I^T \boldsymbol{\varepsilon} + n^{1/2} \mathbf{S}_{11n}^{-1} \Delta_{(\mathbf{X}_I, \mathbf{Y})} - n^{1/2} \mathbf{S}_{11n}^{-1} \Delta_{\mathbf{X}_I} \beta_{01} + o_p(1) \\ &= \mathbf{S}_{11n}^{-1} \left\{ n^{-1/2} \mathbf{X}_I^T \boldsymbol{\varepsilon} + n^{-1/2} \mathbf{X}_I^T (\hat{\mathbf{Y}} - \mathbf{Y}) - n^{-1/2} \mathbf{X}_I^T (\hat{\mathbf{X}}_I - \mathbf{X}_I) \beta_{01} \right. \\ &\quad \left. + K_1 + K_2 + K_3 \right\} + o_p(1), \end{aligned} \quad (\text{A.8})$$

where  $K_1 = n^{-1/2}(\hat{\mathbf{X}}_I - \mathbf{X}_I)^T(\hat{\mathbf{Y}} - \mathbf{Y})$ ,  $K_2 = n^{-1/2}(\hat{\mathbf{X}}_I - \mathbf{X}_I)^T \boldsymbol{\varepsilon}$ ,  $K_3 = n^{-1/2}(\hat{\mathbf{X}}_I - \mathbf{X}_I)^T (\hat{\mathbf{X}}_I - \mathbf{X}_I) \beta_{01}$ . As  $n \rightarrow \infty$ , it follows from Corollary 1 and condition (C7) that  $K_i = o_p(1)$ ,  $i = 1, 2, 3$ . Therefore, by Lemma 5,

$$n^{1/2}(\hat{\beta}_1^\circ - \beta_{01}) = \boldsymbol{\Omega}_1^{-1} \left\{ n^{-1/2} \mathbf{X}_I^T \boldsymbol{\varepsilon} + n^{-1/2} \mathbf{X}_I^T (\hat{\mathbf{Y}} - \mathbf{Y}) - n^{-1/2} \mathbf{X}_I^T (\hat{\mathbf{X}}_I - \mathbf{X}_I) \beta_{01} \right\} + o_p(1). \quad (\text{A.9})$$

We rewrite

$$n^{1/2} s_n^{-1} \mathbf{b}_n^T (\hat{\beta}_1^\circ - \beta_{01}) \equiv s_n^{-1} (\mathbf{I}_1 + \mathbf{I}_2 - \mathbf{I}_3) + o_p(1), \quad (\text{A.10})$$

where  $\mathbf{I}_1 = n^{-1/2} \mathbf{b}_n^T \boldsymbol{\Omega}_1^{-1} \mathbf{X}_I^T \boldsymbol{\varepsilon}$ ,  $\mathbf{I}_2 = n^{-1/2} \mathbf{b}_n^T \boldsymbol{\Omega}_1^{-1} \mathbf{X}_I^T (\hat{\mathbf{Y}} - \mathbf{Y})$  and  $\mathbf{I}_3 = n^{-1/2} \mathbf{b}_n^T \boldsymbol{\Omega}_1^{-1} \mathbf{X}_I^T (\hat{\mathbf{X}}_I - \mathbf{X}_I) \beta_{01}$ . Note that  $\mathbf{I}_1, \mathbf{I}_2$  and  $\mathbf{I}_3$  are sum of independent identically distributed variables, and these terms are not correlated except the  $\mathbf{I}_2$  and  $\mathbf{I}_3$ . Thus, we need to verify that  $\mathbf{I}_1$  and  $\mathbf{I}_2 - \mathbf{I}_3$  satisfy the conditions of the Lindeberg-Feller central limit theorem. For  $\mathbf{I}_1$ , we let  $Z_{ni} = n^{-1/2} \mathbf{b}_n^T \boldsymbol{\Omega}_1^{-1} \mathbf{X}_{Ii}^T \varepsilon_i$ , note that

$$E(|Z_{ni}|^4) = E(Z_{ni}^T Z_{ni})^2 \leq n^{-2} \sigma^4 (\lambda_{\max}(\boldsymbol{\Omega}_1^{-1}))^2 E \left( \sum_{j=1}^{q_n} X_{Iij}^2 \right)^2 = O(q_n^2/n^2), \quad (\text{A.11})$$

where  $\mathbf{X}_{Ii}$  is the  $i$ th sample of  $\mathbf{X}_I$ ,  $X_{Iij}$  is the  $j$ th component of  $\mathbf{X}_{Ii}$ .

Therefore for any  $\eta > 0$ , as  $n \rightarrow \infty$

$$\begin{aligned} \sum_{i=1}^n E(|Z_{ni}|^2 \mathbf{I}(|Z_{ni}| > \eta)) &\leq n (E(|Z_{ni}|^4))^{1/2} \left( E(\mathbf{I}(|Z_{ni}| > \eta))^2 \right)^{1/2} \\ &\leq n (E(|Z_{ni}|^4))^{1/2} \left( \frac{\sigma^2 q_n (\lambda_{\max}(\boldsymbol{\Omega}_1^{-1}))^2}{n \eta^2} \right)^{1/2} \\ &\leq n (O(q_n^2/n^2))^{1/2} \cdot (O(q_n/n))^{1/2} \rightarrow 0, \end{aligned} \quad (\text{A.12})$$

where the first inequality is due to Cauchy-Schwarz inequality, the last inequality follows from Lemma 5. (A.12) implies that  $I_1$  satisfies Lindeberg condition. Similarly, we can also verify that  $I_2 - I_3$  satisfies the conditions of the Lindeberg-Feller central limit theorem, we omit the details.

Through some calculation, we have that

$$Cov(s_n^{-1}(I_1 + I_2 - I_3)) \rightarrow 1. \quad (\text{A.13})$$

Therefore, by the Lindeberg-Feller central limit theorem,  $s_n^{-1}(I_1 + I_2 - I_3) \xrightarrow{D} N(0, 1)$ . This completes the proofs of part (b).  $\square$

**Proof of Theorem 1** Its proof is similar to the proofs of Theorem 1 in (Dai et al., 2018) and thus is omitted.  $\square$

**Proof of Theorem 2** We decompose  $\hat{\sigma}^2$  as

$$\hat{\sigma}^2 = n^{-1}\hat{\mathbf{Y}}^T\hat{\mathbf{Y}} - 2n^{-1}\hat{\boldsymbol{\beta}}^{*T}\hat{\mathbf{X}}^T\hat{\mathbf{Y}} + n^{-1}\hat{\boldsymbol{\beta}}^{*T}\hat{\mathbf{X}}^T\hat{\mathbf{X}}\hat{\boldsymbol{\beta}}^*. \quad (\text{A.14})$$

We first consider  $n^{-1}\hat{\mathbf{Y}}^T\hat{\mathbf{Y}}$ , it is easy to see that

$$n^{-1}\hat{\mathbf{Y}}^T\hat{\mathbf{Y}} - n^{-1}\mathbf{Y}^T\mathbf{Y} = 2n^{-1}(\hat{\mathbf{Y}} - \mathbf{Y})^T\mathbf{Y} + n^{-1}(\hat{\mathbf{Y}} - \mathbf{Y})^T(\hat{\mathbf{Y}} - \mathbf{Y}) \quad (\text{A.15})$$

Note that

$$\begin{aligned} E\left|n^{-1}(\hat{\mathbf{Y}} - \mathbf{Y})^T\mathbf{Y}\right| &= E\left|n^{-1}\sum_{i=1}^n\left(\frac{\psi(U_i)}{\hat{\psi}(U_i)} - 1\right)Y_i^2\right| \\ &\leq n^{-1}\sum_{i=1}^n\left(E\left|\frac{\psi(U_i)}{\hat{\psi}(U_i)} - 1\right|^2\right)^{1/2} \cdot (E|Y_i^2|)^{1/2} \quad (\text{A.16}) \\ &= O\left(a_{1n}^2p_n^2n^{-1/2}h^{-1}\log n\right) \rightarrow 0, \end{aligned}$$

where the first inequality is due to Cauchy-Schwarz inequality, the last equality follows from Lemma 3 and the fact that  $E(Y^4) = O(a_{1n}^4p_n^4)$ . Thus,  $n^{-1}(\hat{\mathbf{Y}} - \mathbf{Y})^T\mathbf{Y} = o_p(1)$ .

Similar to (A.16), we obtain  $n^{-1}(\hat{\mathbf{Y}} - \mathbf{Y})^T(\hat{\mathbf{Y}} - \mathbf{Y}) = o_p(1)$ . This, combined with (A.15)-(A.16) gives

$$n^{-1}\hat{\mathbf{Y}}^T\hat{\mathbf{Y}} - n^{-1}\mathbf{Y}^T\mathbf{Y} = o_p(1). \quad (\text{A.17})$$

Next, we consider  $n^{-1}\hat{\boldsymbol{\beta}}^{*T}\hat{\mathbf{X}}^T\hat{\mathbf{Y}}$ , write

$$n^{-1}\hat{\boldsymbol{\beta}}^{*T}\hat{\mathbf{X}}^T\hat{\mathbf{Y}} - \boldsymbol{\beta}_{0n}^Tn^{-1}\mathbf{X}^T\mathbf{Y} = J_1 + J_2 + J_3, \quad (\text{A.18})$$

where  $J_1 = (\hat{\beta}^* - \beta_{0n})^T \Delta_{(\mathbf{X}, \mathbf{Y})}$ ,  $J_2 = (\hat{\beta}^* - \beta_{0n})^T n^{-1} \mathbf{X}^T \mathbf{Y}$  and  $J_3 = \beta_{0n}^T \Delta_{(\mathbf{X}, \mathbf{Y})}$  with  $\Delta_{(\mathbf{X}, \mathbf{Y})}$  defined as in the proof of Proposition 1. It follows Corollary 1 and Lemma 2 that

$$|J_1| \leq \|\hat{\beta}^* - \beta_{0n}\| \|\Delta_{(\mathbf{X}, \mathbf{Y})}\| = O_p\left(\delta_n \sqrt{q_n^2/n}\right) \cdot O_p\left(a_{1n} p_n n^{-1/2}\right) \rightarrow 0.$$

Similarly,  $|J_2| = o_p(1)$  and  $|J_3| = o_p(1)$ . This, together with (A.18), implies that

$$n^{-1} \hat{\beta}^* \hat{\mathbf{X}}^T \hat{\mathbf{Y}} - n^{-1} \beta_{0n}^T \mathbf{X}^T \mathbf{Y} = o_p(1). \quad (\text{A.19})$$

Similar to the derivation of (A.17) and (A.19), we can obtain that

$$n^{-1} \hat{\beta}^{*T} \hat{\mathbf{X}}^T \hat{\mathbf{X}} \hat{\beta}^* - \beta_{0n}^T \mathbf{S}_n \beta_{0n} = o_p(1). \quad (\text{A.20})$$

By some simple calculations, we can prove that  $n^{-1} \mathbf{Y}^T \mathbf{Y} - E(Y^2) = o_p(1)$ ,  $n^{-1} \beta_{0n}^T \mathbf{X}^T \mathbf{Y} - \beta_{0n}^T E(XY) = o_p(1)$  and  $\beta_{0n}^T \mathbf{S}_n \beta_{0n} - \beta_{0n}^T E(XX^T) \beta_{0n} = o_p(1)$ , where  $X = (X_1, X_2, \dots, X_{p_n})^T$ . This, combined with (A.17), (A.19) and (A.20) yields that

$$\begin{aligned} n^{-1} \hat{\mathbf{Y}}^T \hat{\mathbf{Y}} - E(Y^2) &= o_p(1), \\ n^{-1} \hat{\beta}^* \hat{\mathbf{X}}^T \hat{\mathbf{Y}} - \beta_{0n}^T E(XY) &= o_p(1), \\ n^{-1} \hat{\beta}^{*T} \hat{\mathbf{X}}^T \hat{\mathbf{X}} \hat{\beta}^* - \beta_{0n}^T E(XX^T) \beta_{0n} &= o_p(1). \end{aligned} \quad (\text{A.21})$$

Therefore, the estimator  $\hat{\sigma}^2$  obtained by CBAR is a consistent estimator of  $\sigma^2$ . We complete the proof.  $\square$

**Proof of Theorem 3** Let

$$\begin{aligned} \hat{\beta}^{(l+1)} &= \arg \min_{\beta} \left\{ Q(\beta | \hat{\beta}^{(l)}) \right\} \\ &= \arg \min_{\beta} \left\{ \left\| \hat{\mathbf{Y}} - \hat{\mathbf{X}} \beta \right\|^2 + \lambda_n \sum_{j=1}^{p_n} \frac{\beta_j^2}{\left( \hat{\beta}_j^{(l)} \right)^2} \right\} \end{aligned}$$

and  $\hat{\varepsilon}^{(l+1)} = \hat{\mathbf{Y}} - \hat{\mathbf{X}} \hat{\beta}^{(l+1)}$ . For any  $\hat{\beta}_j \neq 0, j \in \{1, 2, \dots, p_n\}$ , we have that

$$-2 \hat{\mathbf{X}}_j^T \left( \hat{\mathbf{Y}} - \hat{\mathbf{X}} \hat{\beta}^{(l+1)} \right) + 2 \lambda_n \frac{\hat{\beta}_j^{(l+1)}}{\left( \hat{\beta}_j^{(l)} \right)^2} = 0.$$

Thus

$$\frac{\hat{\beta}_j^{(l+1)}}{(\hat{\beta}_j^{(l)})^2} = \frac{\hat{\mathbf{X}}_j^T \hat{\boldsymbol{\varepsilon}}^{(l+1)}}{\lambda_n}. \quad (\text{A.22})$$

Furthermore,

$$\|\hat{\boldsymbol{\varepsilon}}^{(l+1)}\|^2 + \lambda_n \sum_{i=1}^{p_n} \frac{(\hat{\beta}_i^{(l+1)})^2}{(\hat{\beta}_i^{(l)})^2} = Q(\hat{\boldsymbol{\beta}}^{(l+1)} | \hat{\boldsymbol{\beta}}^{(l)}) \leq Q(0 | \hat{\boldsymbol{\beta}}^{(l)}) = \|\hat{\mathbf{Y}}\|^2.$$

Therefore,

$$\|\hat{\boldsymbol{\varepsilon}}^{(l+1)}\| \leq \|\hat{\mathbf{Y}}\|. \quad (\text{A.23})$$

Since  $\lim_{l \rightarrow \infty} \hat{\boldsymbol{\beta}}_j^{(l+1)} = \lim_{l \rightarrow \infty} \hat{\boldsymbol{\beta}}_j^{(l)} = \hat{\boldsymbol{\beta}}_j^*$ . Letting  $l \rightarrow \infty$  on both sides of (A.22), we have that with probability tending to 1

$$\frac{1}{\hat{\beta}_i^*} = \frac{\hat{\mathbf{X}}_i^T (\hat{\mathbf{Y}} - \hat{\mathbf{X}} \hat{\boldsymbol{\beta}}^*)}{\lambda_n}, \quad \frac{1}{\hat{\beta}_j^*} = \frac{\hat{\mathbf{X}}_j^T (\hat{\mathbf{Y}} - \hat{\mathbf{X}} \hat{\boldsymbol{\beta}}^*)}{\lambda_n}$$

for any  $i, j \in \{1, \dots, p_n\}$ . Write  $\hat{\boldsymbol{\varepsilon}}^* = \hat{\mathbf{Y}} - \hat{\mathbf{X}} \hat{\boldsymbol{\beta}}^*$ , we have

$$\left| \frac{1}{\hat{\beta}_i^*} - \frac{1}{\hat{\beta}_j^*} \right| \leq \frac{1}{\lambda_n} \|\hat{\boldsymbol{\varepsilon}}^*\| \|\hat{\mathbf{X}}_i - \hat{\mathbf{X}}_j\| \leq \frac{1}{\lambda_n} \|\hat{\mathbf{Y}}\| \sqrt{2(1 - \rho_{ij})}.$$

We complete the proof.  $\square$

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s42952-026-00372-4>.

**Acknowledgements** The authors thank two anonymous reviewers and the editors for their constructive and insightful comments that have led to significant improvements of the paper. This work was supported by Beijing Natural Science Foundation under Grant No. 4242011 and National Natural Science Foundation of China under Grant No. 62171041.

## Declarations

**Conflict of interest** No potential conflict of interest was reported by the authors.

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